Biomath Booklet

Chapters 3 – 4

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Each section is split into topics. Every topic has its key formulas, then **Lecture Problems**, then **Textbook Problems**. Worked answers are at the end of each section.

3.1 Derivative Formulas for Powers and Polynomials

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3.1.1 Constant, Linear and Constant Multiple

Constant	If $f(x) = k$, then $f'(x) = 0$
Linear	If $f(x) = b + mx$, then $f'(x) = m$ (slope)
Constant multiple	$\frac{d}{dx}\big(cf(x)\big) = cf'(x)$

Lecture Problems

Ex 1 Find $\frac{d}{dx}(5)$.

Ex 2 Find $\frac{d}{dx}(2x + 3)$.

Ex 3 Let $f(x) = 5 + 7x$. Find $\frac{d}{dx}\big(2f(x)\big)$.

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

1 $y = 5$

2 $y = 3x$

8 $y = 5x + 13$

29 $f(x) = kx^2$

35 $w = 3ab^2q$

36 $h(x) = \dfrac{ax + b}{c}$

3.1.2 The Power Rule

Power rule $\dfrac{d}{dx}\left(x^n\right) = nx^{n-1}$

Lecture Problems

Ex 4 Use the power rule to find:

i. $\dfrac{d}{dx}\left(x^3\right)$

ii. $\dfrac{d}{dx}\left(x^{-5}\right)$

iii. $\dfrac{d}{dx}\left(x^{3/2}\right)$

Ex 5 Find the derivative of

i. $f(x) = x^8$

ii. $h(t) = t^7$

iii. $g(x) = \sqrt{x}$

iv. $w(t) = \dfrac{1}{t^3}$

v. $R(x) = 2x^{4.5}$

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

3 $y = x^{12}$

4 $y = x^{-12}$

5 $y = x^{4/3}$

6 $y = 8t^3$

9 $f(x) = \dfrac{1}{x^4}$

17 $g(t) = \dfrac{1}{t^5}$

18 $f(z) = -\dfrac{1}{z^{6.1}}$

19 $y = \dfrac{1}{r^{7/2}}$

20 $y = \sqrt{x}$

21 $h(\theta) = \dfrac{1}{\sqrt[3]{\theta}}$

22 $f(x) = \sqrt{\dfrac{1}{x^3}}$

62 Show that for any power function $f(x) = x^n$, we have $f'(1) = n$.

3.1.3 Derivatives of Polynomials

Sum / difference	$\dfrac{d}{dx}\big(f(x) \pm g(x)\big) = f'(x) \pm g'(x)$
Power rule	$\dfrac{d}{dx}\left(x^n\right) = nx^{n-1}$

Lecture Problems

Ex 6 Differentiate

i. $r(p) = p^5 + p^3$

ii. $f(x) = 5x^2 - 7x^3$

iii. $g(t) = \dfrac{t^2}{4} + 3$

Textbook Problems

Find the derivative. Assume a, b, c, k are constants.

7 $y = 3t^4 - 2t^2$

10 $f(q) = q^3 + 10$

11 $y = x^2 + 5x + 9$

12 $y = 6x^3 + 4x^2 - 2x$

13 $y = 3x^2 + 7x - 9$

14 $y = 8t^3 - 4t^2 + 12t - 3$

15 $y = 4.2q^2 - 0.5q + 11.27$

16 $y = -3x^4 - 4x^3 - 6x + 2$

23 $y = 3t^5 - 5\sqrt{t} + \dfrac{7}{t}$

24 $y = z^2 + \dfrac{1}{2z}$

25 $y = 3t^2 + \dfrac{12}{\sqrt{t}} - \dfrac{1}{t^2}$

26 $h(t) = \dfrac{3}{t} + \dfrac{4}{t^2}$

27 $y = \sqrt{x}\,(x + 1)$

28 $h(\theta) = \theta\left(\theta^{-1/2} - \theta^{-2}\right)$

30 $y = ax^2 + bx + c$

31 $Q = aP^2 + bP^3$

32 $v = at^2 + \dfrac{b}{t^2}$

33 $P = a + b\sqrt{t}$

34 $V = \frac{4}{3}\pi r^2 b$

54 Suppose W is proportional to r^3. The derivative dW/dr is proportional to what power of r?

38 Let $f(x) = x^3 - 4x^2 + 7x - 11$. Find $f'(0)$, $f'(2)$, $f'(-1)$.

3.1.4 Equation of the Tangent Line

Tangent line $y - y_1 = m(x - x_1)$, where $m = f'(x_1)$

Lecture Problems

Ex 7 Find an equation for the tangent line at $x = 1$ to the graph of $y = x^3 + 2x^2 - 5x + 7$. Sketch the graph of the curve and its tangent line on the same axes.

Textbook Problems

49 Find the equation of the line tangent to the graph of f at $(1, 1)$, where f is given by $f(x) = 2x^3 - 2x^2 + 1$.

50

(a) Find the equation of the tangent line to $f(x) = x^3$ at the point where $x = 2$.

(b) Graph the tangent line and the function on the same axes. If the tangent line is used to estimate values of the function, will the estimates be overestimates or underestimates?

51 Find the equation of the line tangent to the graph of $f(t) = 6t - t^2$ at $t = 4$. Sketch the graph of $f(t)$ and the tangent line on the same axes.

3.1.5 Rate of Change and Applications

Rate of change	Rate of change at $t = a$: $f'(a)$
Relative rate	Relative rate of change at $t = a$: $\dfrac{f'(a)}{f(a)} \times 100\%$

Lecture Problems

Ex 8 Find the rate of change of a population of size $P(t) = t^3 + 4t + 1$ at time $t = 2$.

Ex 9 Experiments indicate that when a flea jumps, its height (in meters) after t seconds is given by the function $H(t) = 4.4t - 4.9t^2$.

i. Find $H'(t)$.

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ii. At what rate is $H(t)$ changing after 1 second?

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iii. At what time is $H'(t) = 0$? What is the significance of this time?

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iv. When does the flea "land" (return to the initial height)? At what rate is $H(t)$ changing at this time? Is it increasing or decreasing?

Textbook Problems

40 Find the rate of change of a population of size $P(t) = t^3 + 4t + 1$ at time $t = 2$.

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37 Let $P(q) = 6q - q^2$.

(a) Use a graph of $P(q)$ to determine whether each of the following derivatives is positive, negative, or zero: $P'(1)$, $P'(3)$, $P'(4)$. Explain.

(b) Find $P'(q)$ and the three derivatives in part (a).

39 Let $f(t) = t^2 - 4t + 5$.

(a) Find $f'(t)$.

(b) Find $f'(1)$ and $f'(2)$.

(c) Use a graph of $f(t)$ to check that your answers to part (b) are reasonable. Explain.

61 Let $f(x) = x^3 - 6x^2 - 15x + 20$. Find $f'(x)$ and all values of x for which $f'(x) = 0$. Explain the relationship between these values of x and the graph of $f(x)$.

41 The height of a sand dune (in centimeters) is represented by $f(t) = 700 - 3t^2$, where t is measured in years since 2005. Find $f(5)$ and $f'(5)$. Using units, explain what each means in terms of the sand dune.

42 Zebra mussels are freshwater shellfish that first appeared in the St. Lawrence River in the early 1980s and have spread throughout the Great Lakes. Suppose that t months after they appeared in a small bay, the number of zebra mussels is given by $Z(t) = 300t^2$. How many zebra mussels are in the bay after four months? At what rate is the population growing at that time? Give units.

43 The quantity, Q, in tons, of material at a municipal waste site is a function of the number of years since 2000, with $Q = f(t) = 3t^2 + 100$. Find $f(10)$, $f'(10)$, and the relative rate of change f'/f at $t = 10$. Interpret your answers in terms of waste.

44 The number, N, of acres of harvested land in a region is given by $N = f(t) = 120\sqrt{t}$, where t is the number of years since farming began in the region. Find $f(9)$, $f'(9)$, and the relative rate of change f'/f at $t = 9$. Interpret your answers in terms of harvested land.

47 The time, T, in seconds for one complete oscillation of a pendulum is given by $T = f(L) = 1.111\sqrt{L}$, where L is the length of the pendulum in feet. Find the following quantities, with units, and interpret in terms of the pendulum.

(a) $f(100)$

(b) $f'(100)$

48 Kleiber's Law states that the daily calorie requirement, $C(w)$, of a mammal is proportional to the mammal's body weight w raised to the 0.75 power. If body weight is measured in pounds, the constant of proportionality is approximately 42.

(a) Give formulas for $C(w)$ and $C'(w)$.

(b) Find and interpret (i) $C(10)$ and $C'(10)$ (ii) $C(100)$ and $C'(100)$ (iii) $C(1000)$ and $C'(1000)$

52 If you are outdoors, the wind may make it feel a lot colder than the thermometer reads. You feel the windchill temperature, which, if the air temperature is 20°F, is given in $^\circ$F by $W(v) = 48.17 - 27.2v^{0.16}$, where v is the wind velocity in mph for $5 \leq v \leq 60$.

(a) If the air temperature is 20°F, and the wind is blowing at 40 mph, what is the windchill temperature, to the nearest degree?

(b) Find $W'(40)$, and explain what this means in terms of windchill.

53

(a) Use the formula for the area of a circle of radius r, $A = \pi r^2$, to find dA/dr.

(b) The result from part (a) should look familiar. What does dA/dr represent geometrically?

(c) Use the difference quotient to explain the observation you made in part (b).

55 The cost to produce q items is $C(q) = 1000 + 2q^2$ dollars. Find the marginal cost of producing the 25th item. Interpret your answer in terms of costs.

56 The demand curve for a product is given by $q = 300 - 3p$, where p is the price of the product and q is the quantity that consumers buy at this price.

(a) Write the revenue as a function, $R(p)$, of price.

(b) Find $R'(10)$ and interpret your answer in terms of revenue.

(c) For what prices is $R'(p)$ positive? For what prices is it negative?

57 The yield, Y, of an apple orchard (measured in bushels of apples per acre) is a function of the amount x of fertilizer in pounds used per acre. Suppose $Y = f(x) = 320 + 140x - 10x^2$.

(a) What is the yield if 5 pounds of fertilizer is used per acre?

(b) Find $f'(5)$. Give units with your answer and interpret it in terms of apples and fertilizer.

(c) Given your answer to part (b), should more or less fertilizer be used? Explain.

58 The demand for a product is given, for $p, q \geq 0$, by $p = f(q) = 50 - 0.03q^2$.

(a) Find the p- and q-intercepts for this function and interpret them in terms of demand for this product.

(b) Find $f(20)$ and give units with your answer. Explain what it tells you in terms of demand.

(c) Find $f'(20)$ and give units with your answer. Explain what it tells you in terms of demand.

59 The cost (in dollars) of producing q items is given by $C(q) = 0.08q^3 + 75q + 1000$.

(a) Find the marginal cost function.

(b) Find $C(50)$ and $C'(50)$. Give units with your answers and explain what each is telling you about costs of production.

60 A ball is dropped from the top of the Empire State Building. The height, y, of the ball above the ground (in feet) is given as a function of time, t, (in seconds) by $y = 1250 - 16t^2$.

(a) Find the velocity of the ball at time t. What is the sign of the velocity? Why is this to be expected?

(b) When does the ball hit the ground, and how fast is it going at that time? Give your answer in feet per second and in miles per hour (1 ft/sec $= 15/22$ mph).

63 If the demand curve is a line, we can write $p = b + mq$, where p is the price of the product, q is the quantity sold at that price, and b and m are constants.

(a) Write the revenue as a function of quantity sold.

(b) Find the marginal revenue function.

3.1.6 The Second Derivative

Concavity $f'' > 0 \Rightarrow$ concave up $\cup$ $f'' < 0 \Rightarrow$ concave down $\cap$

Lecture Problems

Ex 10 Find and interpret the second derivatives of

i. $f(x) = x^2$

ii. $g(x) = x^3$

Textbook Problems

45 If $f(t) = 2t^3 - 4t^2 + 3t - 1$, find $f'(t)$ and $f''(t)$.

46 If $f(t) = t^4 - 3t^2 + 5t$, find $f'(t)$ and $f''(t)$.

Answers · Section 3.1

3.1.1 Constant, Linear and Constant Multiple

Ex 1 $\dfrac{d}{dx}(5) = 0$ (derivative of a constant is 0).

Ex 2 $\dfrac{d}{dx}(2x+3) = 2$ (linear function: derivative = slope).

Ex 3 $f'(x) = 7$

$\dfrac{d}{dx}\big(2f(x)\big) = 2\dfrac{d}{dx}\big(f(x)\big) = 2f'(x) = 2(7) = 14$

Another solution: $\dfrac{d}{dx}\big(2(5+7x)\big) = \dfrac{d}{dx}(10+14x) = 14$

1 $y' = 0$

2 $y' = 3$

8 $y' = 5$

29 $f'(x) = 2kx$

35 $\dfrac{dw}{dq} = 3ab^2$

36 $h(x) = \frac{a}{c}x + \frac{b}{c} \Rightarrow h'(x) = \dfrac{a}{c}$

3.1.2 The Power Rule

Ex 4 i. $3x^{3-1} = 3x^2$ ii. $-5x^{-5-1} = -5x^{-6}$ iii. $\frac{3}{2}x^{\frac{3}{2}-1} = \frac{3}{2}x^{1/2}$

Ex 5 i. $f'(x) = 8x^{8-1} = 8x^7$

ii. $h'(t) = 7t^{7-1} = 7t^6$

iii. $g(x) = x^{1/2} \Rightarrow g'(x) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-1/2} = \dfrac{1}{2\sqrt{x}}$

iv. $w(t) = t^{-3} \Rightarrow w'(t) = -3t^{-3-1} = -3t^{-4} = -\dfrac{3}{t^4}$

v. $R'(x) = 2(4.5)x^{4.5-1} = 9x^{3.5}$

3 $y' = 12x^{11}$

4 $y' = -12x^{-13}$

5 $y' = \frac{4}{3}x^{1/3}$

6 $y' = 24t^2$

9 $f(x) = x^{-4} \Rightarrow f'(x) = -4x^{-5} = -\dfrac{4}{x^5}$

17 $g(t) = t^{-5} \Rightarrow g'(t) = -5t^{-6}$

18 $f(z) = -z^{-6.1} \Rightarrow f'(z) = 6.1z^{-7.1}$

19 $y = r^{-7/2} \Rightarrow y' = -\frac{7}{2}r^{-9/2}$

20 $y = x^{1/2} \Rightarrow y' = \frac{1}{2}x^{-1/2} = \dfrac{1}{2\sqrt{x}}$

21 $h = \theta^{-1/3} \Rightarrow h'(\theta) = -\frac{1}{3}\theta^{-4/3}$

22 $f(x) = x^{-3/2} \Rightarrow f'(x) = -\frac{3}{2}x^{-5/2}$

62 $f'(x) = nx^{n-1} \Rightarrow f'(1) = n(1)^{n-1} = n$ (since 1 to any power is 1).

3.1.3 Derivatives of Polynomials

Ex 6 i. $r'(p) = 5p^4 + 3p^2$

ii. $f'(x) = 5(2)x^{2-1} - 7(3)x^{3-1} = 10x - 21x^2$

iii. $g(t) = \frac{1}{4}t^2 + 3 \Rightarrow g'(t) = \frac{1}{4}(2)t^{2-1} + 0 = \frac{1}{2}t = \dfrac{t}{2}$

7 $y' = 12t^3 - 4t$

10 $f'(q) = 3q^2$

11 $y' = 2x + 5$

12 $y' = 18x^2 + 8x - 2$

13 $y' = 6x + 7$

14 $y' = 24t^2 - 8t + 12$

15 $y' = 8.4q - 0.5$

16 $y' = -12x^3 - 12x^2 - 6$

23 $y = 3t^5 - 5t^{1/2} + 7t^{-1} \Rightarrow y' = 15t^4 - \frac{5}{2}t^{-1/2} - 7t^{-2}$

24 $y = z^2 + \frac{1}{2}z^{-1} \Rightarrow y' = 2z - \frac{1}{2}z^{-2}$

25 $y = 3t^2 + 12t^{-1/2} - t^{-2} \Rightarrow y' = 6t - 6t^{-3/2} + 2t^{-3}$

26 $h = 3t^{-1} + 4t^{-2} \Rightarrow h'(t) = -3t^{-2} - 8t^{-3}$

27 $y = x^{3/2} + x^{1/2} \Rightarrow y' = \frac{3}{2}x^{1/2} + \frac{1}{2}x^{-1/2}$

28 $h = \theta^{1/2} - \theta^{-1} \Rightarrow h'(\theta) = \frac{1}{2}\theta^{-1/2} + \theta^{-2}$

30 $y' = 2ax + b$

31 $\dfrac{dQ}{dP} = 2aP + 3bP^2$

32 $v = at^2 + bt^{-2} \Rightarrow \dfrac{dv}{dt} = 2at - 2bt^{-3}$

33 $P = a + bt^{1/2} \Rightarrow \dfrac{dP}{dt} = \frac{b}{2}t^{-1/2}$

34 $\dfrac{dV}{dr} = \frac{8}{3}\pi rb$

54 $W = kr^3 \Rightarrow \dfrac{dW}{dr} = 3kr^2$, so dW/dr is proportional to r^2.

38 $f'(x) = 3x^2 - 8x + 7$

$f'(0) = 7, \quad f'(2) = 3(4) - 8(2) + 7 = 3, \quad f'(-1) = 3(1) + 8 + 7 = 18$

3.1.4 Equation of the Tangent Line

Ex 7 **Find y_1 at $x_1 = 1$:** $y = (1)^3 + 2(1)^2 - 5(1) + 7 = 5 \Rightarrow P(x_1, y_1) = (1, 5)$

Find slope m: $y' = 3x^2 + 4x - 5 \Rightarrow m = y'(1) = 3(1)^2 + 4(1) - 5 = 2$

Eq. of tangent: $y - y_1 = m(x - x_1) \Rightarrow y - 5 = 2(x - 1) \Rightarrow y - 5 = 2x - 2$

$\boxed{y = 2x + 3}$

49 $P(x_1, y_1) = (1, 1) \quad f'(x) = 6x^2 - 4x \Rightarrow m = f'(1) = 2$

$y - 1 = 2(x - 1) \Rightarrow \boxed{y = 2x - 1}$

50 (a) $y_1 = f(2) = 8$, $f'(x) = 3x^2 \Rightarrow m = f'(2) = 12$. $y - 8 = 12(x - 2) \Rightarrow \boxed{y = 12x - 16}$

(b) Near $x = 2$ the graph of x^3 is concave up, so it lies above the tangent line. The estimates are **underestimates**.

51 $f(4) = 24 - 16 = 8 \Rightarrow P = (4, 8)$ $f'(t) = 6 - 2t \Rightarrow m = f'(4) = -2$
$y - 8 = -2(t - 4) \Rightarrow \boxed{y = -2t + 16}$ (parabola opening down, vertex at $t = 3$; the tangent line touches it at $(4, 8)$ and lies above it).

3.1.5 Rate of Change and Applications

Ex 8 ARC $= P'(2)$: $P'(t) = 3t^2 + 4 \Rightarrow P'(2) = 3(2)^2 + 4 = 16$
Relative rate of change: $\text{RRC} = \dfrac{P'(2)}{P(2)} = \dfrac{16}{(2)^3 + 4(2) + 1} = \dfrac{16}{17} \times 100\% \approx 94.1\%$

Ex 9 i. $H'(t) = 4.4 - 4.9(2)t \Rightarrow H'(t) = 4.4 - 9.8t$
ii. $H'(1) = 4.4 - 9.8(1) = -5.4$ m/s
iii. $H'(t) = 0 \Rightarrow 4.4 - 9.8t = 0 \Rightarrow -9.8t = -4.4 \Rightarrow t \approx 0.45$ seconds (critical point). At $t = 0.45$ the flea reaches its maximum height, then the height starts to decrease ($H' > 0$ before, $H' < 0$ after).
iv. $H(t) = 0 \Rightarrow 4.4t - 4.9t^2 = 0 \Rightarrow t(4.4 - 4.9t) = 0 \Rightarrow t = 0$ or $t = \dfrac{4.4}{4.9} \approx 0.898$ seconds.
$H'(0.898) = 4.4 - 9.8(0.898) \approx -4.4$ m/s. The height is decreasing at 0.898 seconds.

40 $P'(t) = 3t^2 + 4 \Rightarrow P'(2) = 3(2)^2 + 4 = 16$. *(Same as lecture Ex 8.)*

37 (a) The graph is a parabola opening down with vertex at $q = 3$. At $q = 1$ the graph is increasing $\Rightarrow P'(1) > 0$; at $q = 3$ it is the top (horizontal tangent) $\Rightarrow P'(3) = 0$; at $q = 4$ it is decreasing $\Rightarrow P'(4) < 0$.
(b) $P'(q) = 6 - 2q$: $P'(1) = 4$, $P'(3) = 0$, $P'(4) = -2$.

39 (a) $f'(t) = 2t - 4$ (b) $f'(1) = -2$, $f'(2) = 0$
(c) The graph is a parabola opening up with vertex at $t = 2$. At $t = 1$ the graph is decreasing (negative slope) and at $t = 2$ it is at the bottom (horizontal tangent, slope 0). So the answers are reasonable.

61 $f'(x) = 3x^2 - 12x - 15 = 3(x^2 - 4x - 5) = 3(x - 5)(x + 1)$
$f'(x) = 0 \Rightarrow x = -1$ or $x = 5$. At these values the tangent line is horizontal: f has a local maximum at $x = -1$ and a local minimum at $x = 5$.

41 $f(5) = 700 - 3(25) = 625$ cm $\quad f'(t) = -6t \Rightarrow f'(5) = -30$ cm/year
In 2010 the sand dune is 625 cm high, and its height is decreasing at a rate of 30 cm per year.

42 $Z(4) = 300(4)^2 = 4800$ zebra mussels.
$Z'(t) = 600t \Rightarrow Z'(4) = 2400$ zebra mussels per month. After 4 months the population is growing by about 2400 mussels per month.

43 $f(10) = 3(100) + 100 = 400$ tons $\quad f'(t) = 6t \Rightarrow f'(10) = 60$ tons/year
$\dfrac{f'(10)}{f(10)} = \dfrac{60}{400} = 0.15 = 15\%$ per year
In 2010 there are 400 tons of waste at the site, increasing at 60 tons per year, which is 15% per year.

44 $f(9) = 120\sqrt{9} = 360$ acres $\quad f(t) = 120t^{1/2} \Rightarrow f'(t) = 60t^{-1/2} \Rightarrow f'(9) = \dfrac{60}{3} = 20$ acres/year

$\dfrac{f'(9)}{f(9)} = \dfrac{20}{360} \approx 0.0556 = 5.56\%$ per year

After 9 years, 360 acres are harvested, increasing by 20 acres per year, which is about 5.56% per year.

47 (a) $f(100) = 1.111\sqrt{100} = 11.11$ seconds: a pendulum 100 feet long takes 11.11 seconds for one complete oscillation.

(b) $f(L) = 1.111L^{1/2} \Rightarrow f'(L) = 0.5555L^{-1/2} \Rightarrow f'(100) = \dfrac{0.5555}{10} \approx 0.056$ seconds per foot: increasing the length from 100 ft by 1 ft increases the time by about 0.056 seconds.

48 (a) $C(w) = 42w^{0.75} \quad C'(w) = 42(0.75)w^{-0.25} = 31.5w^{-0.25}$

(b) (i) $C(10) \approx 236$ calories; $C'(10) \approx 17.7$ calories per pound

(ii) $C(100) \approx 1328$ calories; $C'(100) \approx 9.96$ calories per pound

(iii) $C(1000) \approx 7469$ calories; $C'(1000) \approx 5.60$ calories per pound

A 10-lb mammal needs about 236 calories a day, and each extra pound adds about 17.7 calories. Heavier mammals need more calories, but each extra pound adds fewer calories.

52 (a) $W(40) = 48.17 - 27.2(40)^{0.16} \approx 48.17 - 49.08 \approx -1^\circ\text{F}$

(b) $W'(v) = -27.2(0.16)v^{-0.84} = -4.352v^{-0.84} \Rightarrow W'(40) \approx -0.20$ °F per mph. When the wind is 40 mph, each extra 1 mph of wind makes it feel about 0.2°F colder.

53 (a) $\dfrac{dA}{dr} = 2\pi r$ (b) $2\pi r$ is the **circumference** of the circle.

(c) $\dfrac{A(r+h) - A(r)}{h}$: when the radius grows by a small h, the extra area is a thin ring of length about $2\pi r$ and width h, so the extra area $\approx 2\pi r \cdot h$ and the quotient $\approx 2\pi r$.

55 $C'(q) = 4q \Rightarrow C'(25) = 100$ dollars per item. Producing the 25th item costs about \$100.

56 (a) $R(p) = p \cdot q = p(300 - 3p) = 300p - 3p^2$

(b) $R'(p) = 300 - 6p \Rightarrow R'(10) = 240$. When the price is \$10, raising the price by \$1 increases revenue by about \$240.

(c) $R'(p) > 0$ when $p < 50$; $R'(p) < 0$ when $50 < p \le 100$ (since $q \ge 0$ means $p \le 100$).

57 (a) $f(5) = 320 + 700 - 250 = 770$ bushels per acre

(b) $f'(x) = 140 - 20x \Rightarrow f'(5) = 40$ bushels per acre per pound of fertilizer: using 1 more pound increases the yield by about 40 bushels per acre.

(c) Since $f'(5) > 0$, the yield is still increasing, so **more** fertilizer should be used.

58 (a) p-intercept: $q = 0 \Rightarrow p = 50$: at \$50 nobody buys the product. q-intercept: $50 - 0.03q^2 = 0 \Rightarrow q^2 \approx 1666.7 \Rightarrow q \approx 40.8$: about 41 items are wanted if the product is free.

(b) $f(20) = 50 - 0.03(400) = 38$ dollars: to sell 20 items, the price must be \$38.

(c) $f'(q) = -0.06q \Rightarrow f'(20) = -1.2$ dollars per item: to sell one more item (the 21st), the price must drop by about \$1.20.

59 (a) $C'(q) = 0.24q^2 + 75$

(b) $C(50) = 0.08(125000) + 3750 + 1000 = 14{,}750$ dollars: producing 50 items costs \$14,750.

$C'(50) = 0.24(2500) + 75 = 675$ dollars per item: the 51st item costs about \$675.

60 (a) $v = y' = -32t$ ft/s. The velocity is negative because the height is decreasing (the ball is falling).
(b) $y = 0 \Rightarrow 16t^2 = 1250 \Rightarrow t^2 = 78.125 \Rightarrow t \approx 8.84$ seconds.
$v(8.84) = -32(8.84) \approx -282.8$ ft/s, so the speed is about 282.8 ft/s $\approx 282.8 \times \frac{15}{22} \approx 192.8$ mph.

63 (a) $R(q) = p \cdot q = (b + mq)q = bq + mq^2$ (b) $R'(q) = b + 2mq$

3.1.6 The Second Derivative

Ex 10 i. $f'(x) = 2x,\ f''(x) = 2$. $f''(x)$ is positive for all x on $(-\infty, \infty) \Rightarrow f(x)$ is concave up on $(-\infty, \infty)$.
ii. $g'(x) = 3x^2,\ g''(x) = 3(2x) = 6x$. $g''(x) = 0 \Rightarrow x = 0$.
Test: $g''(-1) = -6 < 0$ (concave down $\cap$), $g''(1) = 6 > 0$ (concave up $\cup$).
Since $g''(x)$ is negative on $(-\infty, 0)$ and positive on $(0, \infty)$, $g(x)$ is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$.

45 $f'(t) = 6t^2 - 8t + 3 \quad f''(t) = 12t - 8$

46 $f'(t) = 4t^3 - 6t + 5 \quad f''(t) = 12t^2 - 6$

3.2 Exponential and Logarithmic Functions

In this section

1. The Exponential Rule 2 lecture · 22 textbook
2. The Derivative of ln x 1 lecture · 6 textbook
3. Equation of the Tangent Line 1 lecture · 7 textbook
4. Rate of Change and Applications 2 lecture · 15 textbook

3.2.1 The Exponential Rule

Exponential rule $\dfrac{d}{dx}\left(a^x\right) = (\ln a)\, a^x$ and $\dfrac{d}{dx}\left(a^{kx}\right) = k(\ln a)\, a^{kx}$

Base e $\dfrac{d}{dx}\left(e^x\right) = e^x$ and $\dfrac{d}{dx}\left(e^{kx}\right) = ke^{kx}$

Lecture Problems

Ex 1 Differentiate $y = 2.3^x + 5e^x$.

Ex 2 Find the derivative of $P = 5 + 3x^2 - 7e^{-0.2x}$.

Textbook Problems

Differentiate. Assume A, B, and C are constants.

1 $f(x) = 2e^x + x^2$

2 $P = 3t^3 + 2e^t$

3 $y = 5t^2 + 4e^t$

4 $f(x) = x^3 + 3^x$

5 $y = 2^x + \dfrac{2}{x^3}$

6 $y = 5 \cdot 5^t + 6 \cdot 6^t$

7 $f(x) = 2^x + 2 \cdot 3^x$

8 $y = 4 \cdot 10^x - x^3$

9 $y = 3x - 2 \cdot 4^x$

10 $y = 5 \cdot 2^x - 5x + 4$

11 $f(t) = e^{3t}$

12 $y = e^{0.7t}$

13 $y = e^{-4t}$

14 $P = e^{-0.2t}$

15 $P = 50e^{-0.6t}$

16 $P = 200e^{0.12t}$

17 $P(t) = 3000(1.02)^t$

18 $P(t) = 12.41(0.94)^t$

19 $P(t) = Ce^t$

20 $y = B + Ae^t$

21 $f(x) = Ae^x - Bx^2 + C$

22 $y = 10^x + \dfrac{10}{x}$

3.2.2 The Derivative of ln x

Natural log $\dfrac{d}{dx}(\ln x) = \dfrac{1}{x}$

Lecture Problems

Ex 3 Differentiate $y = 5\ln t + 7e^t - 4t^2 + 12$.

Textbook Problems

Differentiate. Assume A, B, and C are constants.

23 $R = 3\ln q$

24 $D = 10 - \ln p$

25 $y = t^2 + 5\ln t$

26 $R(q) = q^2 - 2\ln q$

27 $y = x^2 + 4x - 3\ln x$

28 $f(t) = Ae^t + B\ln t$

3.2.3 Equation of the Tangent Line

Tangent line $y - y_1 = m(x - x_1)$, where $m = f'(x_1)$

Lecture Problems

Ex 6 Find the equation of the tangent line to the graph of $f(x) = \ln x$ at the point where $x = 2$.

Textbook Problems

29 For $f(t) = 4 - 2e^t$, find $f'(-1)$, $f'(0)$, and $f'(1)$. Graph $f(t)$, and draw tangent lines at $t = -1$, $t = 0$, and $t = 1$. Do the slopes of the lines match the derivatives you found?

30 Find the equation of the tangent line to the graph of $y = 3^x$ at $x = 1$. Check your work by sketching a graph of the function and the tangent line on the same axes.

31 Find the equation of the tangent line to $y = e^{-2t}$ at $t = 0$. Check by sketching the graphs of $y = e^{-2t}$ and the tangent line on the same axes.

32 Find the equation of the tangent line to $f(x) = 10e^{-0.2x}$ at $x = 4$.

41 Find the value of c where the line l tangent to the graph of $y = 2^x$ at $(0, 1)$ intersects the x-axis.

49

(a) Find the equation of the tangent line to $y = \ln x$ at $x = 1$.

(b) Use it to calculate approximate values for $\ln(1.1)$ and $\ln(2)$.

(c) Using a graph, explain whether the approximate values are smaller or larger than the true values. Would the same result have held if you had used the tangent line to estimate $\ln(0.9)$ and $\ln(0.5)$? Why?

50 Find the quadratic polynomial $g(x) = ax^2 + bx + c$ which best fits the function $f(x) = e^x$ at $x = 0$, in the sense that $g(0) = f(0)$, and $g'(0) = f'(0)$, and $g''(0) = f''(0)$. Using a computer or calculator, sketch graphs of f and g on the same axes. What do you notice?

3.2.4 Rate of Change and Applications

Rate of change	Rate of change at $t = a$: $f'(a)$
Relative rate	Relative rate of change at $t = a$: $\dfrac{f'(a)}{f(a)} \times 100\%$

Lecture Problems

Ex 4 The population of Nevada, P, in millions, can be approximated by $P = 2.020(1.036)^t$, where t is years since the start of 2000.

i. At what rate was the population growing at the beginning of 2009? Give units with your answer.

ii. What is the relative rate of change (RRC) of the population of Nevada at the beginning of 2009?

Ex 5 It is projected that t years from now, the population of a certain country will become $P(t) = 50e^{0.02t}$ million.

i. At what rate will the population be changing (with respect to time) 10 years from now?

ii. At what percentage (i.e., RRC) will the population be changing (with respect to time) t years from now? Does this percentage rate depend on t or is it constant?

Textbook Problems

33 A fish population is approximated by $P(t) = 10e^{0.6t}$, where t is in months. Calculate and use units to explain what each of the following tells us about the population:

(a) $P(12)$

(b) $P'(12)$

34 The world's population is about $f(t) = 6.8e^{0.012t}$ billion, where t is time in years since 2009. Find $f(0)$, $f'(0)$, $f(10)$, and $f'(10)$. Using units, interpret your answers in terms of population.

35 The demand curve for a product is given by $q = f(p) = 10{,}000e^{-0.25p}$, where q is the quantity sold and p is the price of the product, in dollars. Find $f(2)$ and $f'(2)$. Explain in economic terms what information each of these answers gives you.

36 Worldwide production of solar power, in megawatts, can be modeled by $f(t) = 1040(1.3)^t$, where t is years since 2000. Find $f(0)$, $f'(0)$, $f(15)$, and $f'(15)$. Give units and interpret your answers in terms of solar power.

37 A new DVD is available for sale in a store one week after its release. The cumulative revenue, $\$R$, from sales of the DVD in this store in week t after its release is $R = f(t) = 350 \ln t$ with $t > 1$. Find $f(5)$, $f'(5)$, and the relative rate of change f'/f at $t = 5$. Interpret your answers in terms of revenue.

38 In 2009, the population of Hungary was approximated by $P = 9.906(0.997)^t$, where P is in millions and t is in years since 2009. Assume the trend continues.

(a) What does this model predict for the population of Hungary in the year 2020?

(b) How fast (in people/year) does this model predict Hungary's population will be decreasing in 2020?

39 With t in years since January 1, 2010, the population P of Slim Chance is predicted by $P = 35{,}000(0.98)^t$. At what rate will the population be changing on January 1, 2023?

40 Some antique furniture increased very rapidly in price over the past decade. For example, the price of a particular rocking chair is well approximated by $V = 75(1.35)^t$, where V is in dollars and t is in years since 2000. Find the rate, in dollars per year, at which the price is increasing at time t.

42 At a time t hours after it was administered, the concentration of a drug in the body is $f(t) = 27e^{-0.14t}$ ng/ml. What is the concentration 4 hours after it was administered? At what rate is the concentration changing at that time?

43 The cost of producing a quantity, q, of a product is given by $C(q) = 1000 + 30e^{0.05q}$ dollars. Find the cost and the marginal cost when $q = 50$. Interpret these answers in economic terms.

44 With time, t, in minutes, the temperature, H, in degrees Celsius, of a bottle of water put in the refrigerator at $t = 0$ is given by $H = 4 + 16e^{-0.02t}$. How fast is the water cooling initially? After 10 minutes? Give units.

45 Carbon-14 is a radioactive isotope used to date objects. If A_0 represents the initial amount of carbon-14 in the object, then the quantity remaining at time t, in years, is $A(t) = A_0e^{-0.000121t}$.

(a) A tree, originally containing 185 micrograms of carbon-14, is now 500 years old. At what rate is the carbon-14 decaying?

(b) In 1988, scientists found that the Shroud of Turin, which was reputed to be the burial cloth of Jesus, contained 91% of the amount of carbon-14 in freshly made cloth of the same material. According to this data, how old was the Shroud of Turin in 1988?

46 For the cost function $C = 1000 + 300 \ln q$ (in dollars), find the cost and the marginal cost at a production level of 500. Interpret your answers in economic terms.

47 In 2009, the population, P, of India was 1.166 billion and growing at 1.5% annually.

(a) Give a formula for P in terms of time, t, measured in years since 2009.

(b) Find $\dfrac{dP}{dt}$, $\left.\dfrac{dP}{dt}\right|_{t=0}$, and $\left.\dfrac{dP}{dt}\right|_{t=25}$. What do each of these represent in practical terms?

48 In 2009, the population of Mexico was 111 million and growing 1.13% annually, while the population of the US was 307 million and growing 0.975% annually. If we measure growth rates in people/year, which population was growing faster in 2009?

Answers · Section 3.2

3.2.1 The Exponential Rule

Ex 1 $a = 2.3$: $y' = \ln(2.3)\,2.3^x + 5e^x$

Ex 2 $e^{-0.2x}$: $k = -0.2$

$P' = 0 + 3(2x) - 7(-0.2)e^{-0.2x}$

$\boxed{P' = 6x + 1.4e^{-0.2x}}$

1 $f'(x) = 2e^x + 2x$

2 $P' = 9t^2 + 2e^t$

3 $y' = 10t + 4e^t$

4 $f'(x) = 3x^2 + (\ln 3)3^x$

5 $y = 2^x + 2x^{-3} \Rightarrow y' = (\ln 2)2^x - 6x^{-4}$

6 $y' = 5(\ln 5)5^t + 6(\ln 6)6^t$

7 $f'(x) = (\ln 2)2^x + 2(\ln 3)3^x$

8 $y' = 4(\ln 10)10^x - 3x^2$

9 $y' = 3 - 2(\ln 4)4^x$

10 $y' = 5(\ln 2)2^x - 5$

11 $f'(t) = 3e^{3t}$

12 $y' = 0.7e^{0.7t}$

13 $y' = -4e^{-4t}$

14 $P' = -0.2e^{-0.2t}$

15 $P' = 50(-0.6)e^{-0.6t} = -30e^{-0.6t}$

16 $P' = 200(0.12)e^{0.12t} = 24e^{0.12t}$

17 $P'(t) = 3000\ln(1.02)(1.02)^t \approx 59.41(1.02)^t$

18 $P'(t) = 12.41\ln(0.94)(0.94)^t \approx -0.768(0.94)^t$

19 $P'(t) = Ce^t$

20 $y' = Ae^t$

21 $f'(x) = Ae^x - 2Bx$

22 $y = 10^x + 10x^{-1} \Rightarrow y' = (\ln 10)10^x - \dfrac{10}{x^2}$

3.2.2 The Derivative of ln x

Ex 3 $y' = 5\left(\dfrac{1}{t}\right) + 7e^t - 4(2t) + 0$

$\boxed{y' = \dfrac{5}{t} + 7e^t - 8t}$

23 $R' = \dfrac{3}{q}$

24 $D' = -\dfrac{1}{p}$

25 $y' = 2t + \dfrac{5}{t}$

26 $R'(q) = 2q - \dfrac{2}{q}$

27 $y' = 2x + 4 - \dfrac{3}{x}$

28 $f'(t) = Ae^t + \dfrac{B}{t}$

3.2.3 Equation of the Tangent Line

Ex 6 **Find y_1 when $x_1 = 2$:** $y_1 = f(2) = \ln(2) \Rightarrow (x_1, y_1) = (2, \ln 2)$
Find slope m: $f'(x) = \dfrac{1}{x} \Rightarrow m = f'(2) = \dfrac{1}{2}$
Eq. of tangent: $y - \ln(2) = \frac{1}{2}(x - 2) \Rightarrow y - \ln(2) = \frac{1}{2}x - 1 \Rightarrow y = \frac{1}{2}x - 1 + \ln(2)$
$\boxed{y = \frac{1}{2}x - 0.307}$

29 $f'(t) = -2e^t$: $f'(-1) = -2e^{-1} \approx -0.736$, $f'(0) = -2$, $f'(1) = -2e \approx -5.437$
The graph is decreasing and gets steeper as t increases, so all three tangent lines have negative slopes that get steeper. Yes, the slopes match.

30 $y_1 = 3^1 = 3 \Rightarrow (1, 3)$ $y' = (\ln 3)3^x \Rightarrow m = 3\ln 3 \approx 3.296$
$y - 3 = 3.296(x - 1) \Rightarrow \boxed{y = 3.296x - 0.296}$

31 $y_1 = e^0 = 1 \Rightarrow (0, 1)$ $y' = -2e^{-2t} \Rightarrow m = -2$
$y - 1 = -2(t - 0) \Rightarrow \boxed{y = -2t + 1}$

32 $y_1 = f(4) = 10e^{-0.8} \approx 4.493 \Rightarrow (4,\ 4.493)$
$f'(x) = 10(-0.2)e^{-0.2x} = -2e^{-0.2x} \Rightarrow m = f'(4) = -2e^{-0.8} \approx -0.899$
$y - 4.493 = -0.899(x - 4) \Rightarrow \boxed{y = -0.899x + 8.088}$

41 $y' = (\ln 2)2^x \Rightarrow m = \ln 2$ at $x = 0$. Tangent: $y - 1 = (\ln 2)(x - 0) \Rightarrow y = 1 + (\ln 2)x$
At the x-axis $y = 0$: $0 = 1 + (\ln 2)c \Rightarrow c = -\dfrac{1}{\ln 2} \approx -1.443$

49 (a) $y_1 = \ln 1 = 0$, $y' = \dfrac{1}{x} \Rightarrow m = 1$. $y - 0 = 1(x - 1) \Rightarrow \boxed{y = x - 1}$
(b) $\ln(1.1) \approx 1.1 - 1 = 0.1$ $\ln(2) \approx 2 - 1 = 1$
(c) $\ln x$ is concave down, so its graph lies below the tangent line. The approximations are larger than the true values ($\ln 1.1 \approx 0.0953$, $\ln 2 \approx 0.693$). Yes, the same holds for $\ln(0.9) \approx -0.1$ and $\ln(0.5) \approx -0.5$ (true values -0.105 and -0.693), because the curve is below the tangent line on both sides of $x = 1$.

50 $f(0) = f'(0) = f''(0) = e^0 = 1$. $g'(x) = 2ax + b$, $g''(x) = 2a$
$g(0) = c = 1$, $g'(0) = b = 1$, $g''(0) = 2a = 1 \Rightarrow a = \frac{1}{2}$
$\boxed{g(x) = \frac{1}{2}x^2 + x + 1}$. The two graphs are very close near $x = 0$ and separate as x moves away from 0.

3.2.4 Rate of Change and Applications

Ex 4 In 2000, $t = 0$, so the beginning of 2009 is $t = 9$.
i. $P' = 2.02\ln(1.036)(1.036)^t$
$P'(9) = 2.02\ln(1.036)(1.036)^9 \approx 0.0982$ million/year
The population of Nevada was growing at a rate of about 0.0982 million, or 98,200, people per year at the start of 2009.
ii. $\text{RRC} = \dfrac{P'(9)}{P(9)} = \dfrac{0.0982}{2.020(1.036)^9} \times 100\% \approx 0.0354 \times 100\% = 3.54\%$ per year

Ex 5 i. $P'(t) = 50(0.02)e^{0.02t} = e^{0.02t}$ million/year
$P'(10) = e^{0.02(10)} \approx 1.221$ million/year
10 years from now, the population will be growing by 1.221 million per year.
ii. $\text{RRC} = \dfrac{P'(t)}{P(t)} = \dfrac{e^{0.02t}}{50e^{0.02t}} \times 100\% = \dfrac{1}{50} \times 100\% = 2\%$ per year
The percentage rate is constant and does not depend on t; it is the percentage of k (the continuous rate of change).

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33 (a) $P(12) = 10e^{7.2} \approx 13{,}394$ fish: after 12 months there are about 13,394 fish.
(b) $P'(t) = 10(0.6)e^{0.6t} = 6e^{0.6t} \Rightarrow P'(12) = 6e^{7.2} \approx 8037$ fish per month: after 12 months the population is growing by about 8037 fish per month.

---

34 $f'(t) = 6.8(0.012)e^{0.012t} = 0.0816e^{0.012t}$
$f(0) = 6.8$ billion people in 2009; $f'(0) = 0.0816$ billion people per year: in 2009 the population grows by about 81.6 million people per year.
$f(10) = 6.8e^{0.12} \approx 7.667$ billion people in 2019; $f'(10) = 0.0816e^{0.12} \approx 0.092$ billion people per year: in 2019 it grows by about 92 million people per year.

---

35 $f(2) = 10{,}000e^{-0.5} \approx 6065$ items: at a price of 2 dollars, about 6065 items are sold.
$f'(p) = 10{,}000(-0.25)e^{-0.25p} = -2500e^{-0.25p} \Rightarrow f'(2) = -2500e^{-0.5} \approx -1516$ items per dollar: at a price of 2 dollars, raising the price by 1 dollar decreases sales by about 1516 items.

---

36 $f'(t) = 1040\ln(1.3)(1.3)^t$
$f(0) = 1040$ megawatts in 2000; $f'(0) = 1040\ln(1.3) \approx 272.9$ megawatts per year: production was growing by about 273 MW per year in 2000.
$f(15) = 1040(1.3)^{15} \approx 53{,}233$ megawatts in 2015; $f'(15) \approx 13{,}967$ megawatts per year in 2015.

---

37 $f(5) = 350\ln 5 \approx 563.30$ dollars: by week 5 the revenue is about 563.30 dollars.
$f'(t) = \dfrac{350}{t} \Rightarrow f'(5) = 70$ dollars per week: in week 5 revenue is increasing by about 70 dollars per week.
$\dfrac{f'(5)}{f(5)} = \dfrac{70}{563.30} \approx 0.124 = 12.4\%$ per week.

---

38 2020 means $t = 11$.
(a) $P(11) = 9.906(0.997)^{11} \approx 9.584$ million people.
(b) $P'(t) = 9.906\ln(0.997)(0.997)^t \Rightarrow P'(11) \approx -0.0288$ million people per year. The population is decreasing by about 28,800 people per year.

---

39 January 1, 2023 means $t = 13$.
$P'(t) = 35{,}000\ln(0.98)(0.98)^t \Rightarrow P'(13) \approx -544$ people per year. The population will be decreasing by about 544 people per year.

---

40 $\dfrac{dV}{dt} = 75\ln(1.35)(1.35)^t \approx 22.51(1.35)^t$ dollars per year.

---

42 $f(4) = 27e^{-0.56} \approx 15.42$ ng/ml
$f'(t) = 27(-0.14)e^{-0.14t} = -3.78e^{-0.14t} \Rightarrow f'(4) = -3.78e^{-0.56} \approx -2.16$ ng/ml per hour. After 4 hours the concentration is decreasing by about 2.16 ng/ml per hour.

---

43 $C(50) = 1000 + 30e^{2.5} \approx 1365.47$ dollars: producing 50 items costs about 1365.47 dollars.
$C'(q) = 30(0.05)e^{0.05q} = 1.5e^{0.05q} \Rightarrow C'(50) = 1.5e^{2.5} \approx 18.27$ dollars per item: the 51st item costs about 18.27 dollars.

44 $H'(t) = 16(-0.02)e^{-0.02t} = -0.32e^{-0.02t}$
$H'(0) = -0.32$ °C per minute: initially the water cools by 0.32°C per minute.
$H'(10) = -0.32e^{-0.2} \approx -0.262$ °C per minute: after 10 minutes it cools by about 0.26°C per minute.

45 (a) $A(t) = 185e^{-0.000121t} \Rightarrow A'(t) = 185(-0.000121)e^{-0.000121t}$
$A'(500) \approx -0.0211$ micrograms per year: the carbon-14 is decaying at about 0.021 micrograms per year.
(b) $0.91A_0 = A_0e^{-0.000121t} \Rightarrow \ln(0.91) = -0.000121t \Rightarrow t = \dfrac{\ln(0.91)}{-0.000121} \approx 779$ years old.

46 $C(500) = 1000 + 300\ln 500 \approx 2864.38$ dollars: producing 500 items costs about 2864.38 dollars.
$C'(q) = \dfrac{300}{q} \Rightarrow C'(500) = 0.60$ dollars per item: the 501st item costs about 0.60 dollars.

47 (a) $P = 1.166(1.015)^t$ billion
(b) $\dfrac{dP}{dt} = 1.166\ln(1.015)(1.015)^t \approx 0.01736(1.015)^t$ billion people per year: the rate of growth at time t.
$\left.\dfrac{dP}{dt}\right|_{t=0} \approx 0.0174$ billion per year: in 2009 the population grows by about 17.4 million people per year.
$\left.\dfrac{dP}{dt}\right|_{t=25} \approx 0.0252$ billion per year: in 2034 it grows by about 25.2 million people per year.

48 Mexico: $P = 111(1.0113)^t \Rightarrow P'(0) = 111\ln(1.0113) \approx 1.25$ million people per year.
US: $P = 307(1.00975)^t \Rightarrow P'(0) = 307\ln(1.00975) \approx 2.98$ million people per year.
The US population was growing faster in people per year (even though its percentage rate is smaller).

3.3 The Chain Rule

In this section

1. The Chain Rule for Powers 1 lecture · 15 textbook
2. The Chain Rule for Exponentials and Logarithms 2 lecture · 17 textbook
3. The Chain Rule with Values and Graphs 2 lecture · 8 textbook
4. Relative Rates and Logarithms 2 lecture · 13 textbook

3.3.1 The Chain Rule for Powers

Chain rule $\dfrac{dy}{dt} = \dfrac{dy}{dz} \cdot \dfrac{dz}{dt}$ or $\dfrac{d}{dt}\big(f(g(t))\big) = f'(g(t)) \cdot g'(t)$

Power $\dfrac{d}{dt}\Big(\big(f(t)\big)^n\Big) = n\big(f(t)\big)^{n-1} f'(t)$

Lecture Problems

Ex 1 Find the derivative of the function $y = (4t^2 + 1)^7$.

Textbook Problems

Find the derivative of the function.

1 $(4x^2 + 1)^7$

2 $f(x) = (x + 1)^{99}$

3 $R = (q^2 + 1)^4$

4 $w = (t^2 + 1)^{100}$

5 $w = (t^3 + 1)^{100}$

6 $w = (5r - 6)^3$

7 $y = \sqrt{s^3 + 1}$

9 $C = 12(3q^2 - 5)^3$

22 $Q = 100(t^2 + 5)^{0.5}$

24 $y = (5 + e^x)^2$

25 $P = (1 + \ln x)^{0.5}$

26 $\sqrt{e^x + 1}$

27 $f(x) = \sqrt{1 - x^2}$

28 $f(\theta) = (e^\theta + e^{-\theta})^{-1}$

39 Find the equation of the tangent line to $f(x) = (x - 1)^3$ at the point where $x = 2$.

3.3.2 The Chain Rule for Exponentials and Logarithms

Exponential	$\dfrac{d}{dt}\left(e^{f(t)}\right) = e^{f(t)} f'(t)$
Logarithm	$\dfrac{d}{dt}\big(\ln f(t)\big) = \dfrac{f'(t)}{f(t)}$

Lecture Problems

Ex 2 Find the derivative of each of the following:

i. $(3t^3 - t)^5$

ii. $\ln(q^2 + 1)$

Ex 3 Differentiate each of the following:

i. $5\ln(2t^2+3)$

ii. $\sqrt{1+2e^{5t}}$

iii. $\dfrac{1}{\sqrt[3]{5x^2+7}}$

Textbook Problems

Find the derivative of the function.

8 $y = 12 - 3x^2 + 2e^{3x}$

10 $f(x) = 6e^{5x} + e^{-x^2}$

11 $y = 5e^{5t+1}$

12 $w = e^{-3t^2}$

13 $w = e^{\sqrt{s}}$

14 $y = \ln(5t+1)$

15 $f(x) = \ln(1-x)$

16 $f(t) = \ln(t^2+1)$

17 $f(x) = \ln(1-e^{-x})$

18 $f(x) = \ln(e^x+1)$

19 $f(t) = 5\ln(5t+1)$

20 $g(t) = \ln(4t+9)$

21 $y = 5 + \ln(3t + 2)$

23 $y = 5x + \ln(x + 2)$

40 A firm estimates that the total revenue, R, received from the sale of q goods is given by $R = \ln(1 + 1000q^2)$. Calculate the marginal revenue when $q = 10$.

41 The distance, s, of a moving body from a fixed point is given as a function of time by $s = 20e^{t/2}$. Find the velocity, v, of the body as a function of t.

42 If you invest P dollars in a bank account at an annual interest rate of $r\%$, then after t years you will have B dollars, where $B = P\left(1 + \dfrac{r}{100}\right)^t$.

(a) Find dB/dt, assuming P and r are constant. In terms of money, what does dB/dt represent?

(b) Find dB/dr, assuming P and t are constant. In terms of money, what does dB/dr represent?

3.3.3 The Chain Rule with Values and Graphs

Chain rule $\dfrac{dy}{dt} = \dfrac{dy}{dz} \cdot \dfrac{dz}{dt}$ or $\dfrac{d}{dt}\big(f(g(t))\big) = f'(g(t)) \cdot g'(t)$

Lecture Problems

Ex 4 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Given that $f(1) = 1$, $f'(0) = 0$, $f'(1) = 3$, $g(1) = 0$, $g'(1) = 1$. Evaluate:

i. $h'(1)$

ii. $k'(1)$

Ex 5 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Use Figure 3.17 to estimate:

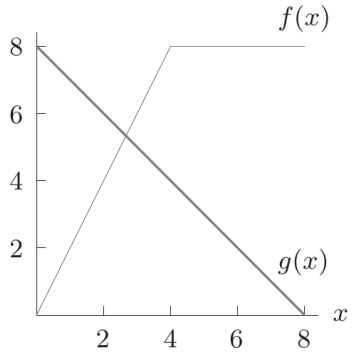


Figure 3.17

i. $h'(1)$

ii. $k'(2)$

Textbook Problems

44 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} f(g(x))\right|_{x=30}$.

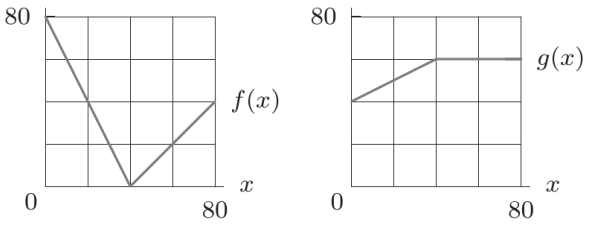


Figure 3.18

45 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} f(g(x))\right|_{x=70}$.

46 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} g(f(x))\right|_{x=30}$.

47 Use Figure 3.18 to evaluate $\left.\frac{d}{dx} g(f(x))\right|_{x=70}$.

48 Let $h(x) = f(g(x))$ and $k(x) = g(f(x))$. Use Figure 3.19 to estimate $h'(1)$.

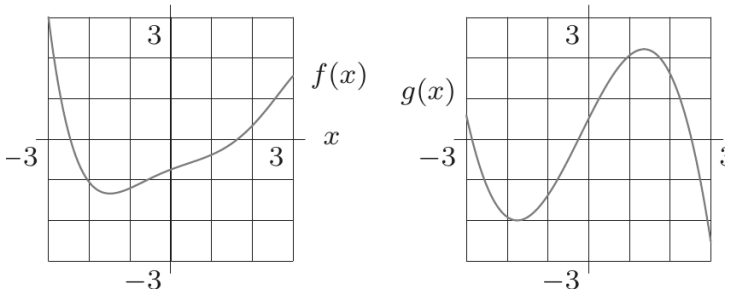


Figure 3.19

49 Use Figure 3.19 to estimate $k'(1)$.

50 Use Figure 3.19 to estimate $h'(2)$.

51 Use Figure 3.19 to estimate $k'(2)$.

3.3.4 Relative Rates and Logarithms

Relative rate Relative rate of change of $f(t) = \dfrac{f'(t)}{f(t)} = \dfrac{d}{dt}\big(\ln f(t)\big)$

Lecture Problems

Ex 6 Find the relative rate of change of the exponential function $z = P_0 e^{kt}$.

Ex 7 The surface area S of a mammal, in cm^2, is a function of the body mass, M, of the mammal, in kilograms, and is given by $S = 1095M^{2/3}$. Find the relative rate of change of S with respect to M and evaluate for a human with body mass 70 kilograms. Interpret your answer.

Textbook Problems

29 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 10t + 5$.

30 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 15t + 12$.

31 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 8e^{5t}$.

32 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 30e^{-7t}$.

33 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 6t^2$.

34 Find the relative rate of change, $f'(t)/f(t)$, of the function $f(t) = 35t^{-4}$.

35 Find the relative rate of change of $f(t) = 5e^{1.5t}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

36 Find the relative rate of change of $f(t) = 6.8e^{-0.5t}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

37 Find the relative rate of change of $f(t) = 3t^2$ using the formula $\dfrac{d}{dt} \ln f(t)$.

38 Find the relative rate of change of $f(t) = 4.5t^{-4}$ using the formula $\dfrac{d}{dt} \ln f(t)$.

43 The distance traveled, D in feet, is a function of time, t, in seconds, with $D = f(t) = \sqrt{t^3 + 1}$. Find $f(10)$, $f'(10)$, and the relative rate of change f'/f at $t = 10$. Interpret your answers in terms of distance traveled.

52 Some economists suggest that an extra year of education increases a person's wages, on average, by about 14%. Assume you could make 10 dollars per hour with your current level of education and that inflation increases wages at a continuous rate of 3.5% per year.

(a) How much would you make per hour with four additional years of education?

(b) What is the difference between your wages in 20 years' time with and without the additional four years of education?

(c) Is the difference you found in part (b) increasing with time? If so, at what rate? (Assume the number of additional years of education stays fixed at four.)

53 Show that if the graphs of $f(t)$ and $h(t) = Ae^{kt}$ are tangent at $t = a$, then k is the relative rate of change of f at $t = a$.

Answers · Section 3.3

3.3.1 The Chain Rule for Powers

Ex 1 $y' = 7(4t^2+1)^{7-1}(8t+0)$

$\boxed{y' = 56t(4t^2+1)^6}$

Using z: $y = z^7,\ z = 4t^2+1,\ \dfrac{dy}{dt} = \dfrac{dy}{dz}\cdot\dfrac{dz}{dt} = 7z^6\cdot(8t) = 7(4t^2+1)^6(8t)$

1 $7(4x^2+1)^6(8x) = 56x(4x^2+1)^6$

2 $f'(x) = 99(x+1)^{98}$

3 $R' = 4(q^2+1)^3(2q) = 8q(q^2+1)^3$

4 $w' = 100(t^2+1)^{99}(2t) = 200t(t^2+1)^{99}$

5 $w' = 100(t^3+1)^{99}(3t^2) = 300t^2(t^3+1)^{99}$

6 $w' = 3(5r-6)^2(5) = 15(5r-6)^2$

7 $y = (s^3+1)^{1/2} \Rightarrow y' = \frac{1}{2}(s^3+1)^{-1/2}(3s^2) = \dfrac{3s^2}{2\sqrt{s^3+1}}$

9 $C' = 12\cdot 3(3q^2-5)^2(6q) = 216q(3q^2-5)^2$

22 $Q' = 100(0.5)(t^2+5)^{-0.5}(2t) = \dfrac{100t}{\sqrt{t^2+5}}$

24 $y' = 2(5+e^x)e^x$

25 $P' = 0.5(1+\ln x)^{-0.5}\cdot\dfrac{1}{x} = \dfrac{0.5}{x\sqrt{1+\ln x}}$

26 $\frac{1}{2}(e^x+1)^{-1/2}e^x = \dfrac{e^x}{2\sqrt{e^x+1}}$

27 $f'(x) = \frac{1}{2}(1-x^2)^{-1/2}(-2x) = \dfrac{-x}{\sqrt{1-x^2}}$

28 $f'(\theta) = -(e^\theta+e^{-\theta})^{-2}(e^\theta-e^{-\theta}) = -\dfrac{e^\theta-e^{-\theta}}{(e^\theta+e^{-\theta})^2}$

39 $y_1 = f(2) = (1)^3 = 1 \Rightarrow (2,1)\quad f'(x) = 3(x-1)^2 \Rightarrow m = f'(2) = 3$

$y-1 = 3(x-2) \Rightarrow \boxed{y = 3x-5}$

3.3.2 The Chain Rule for Exponentials and Logarithms

Ex 2 i. $y' = 5(3t^3-t)^4(9t^2-1) = \boxed{5(9t^2-1)(3t^3-t)^4}$

ii. $y' = \dfrac{1}{q^2+1}(2q+0) = \boxed{\dfrac{2q}{q^2+1}}$

Ex 3 i. $w' = 5\cdot\dfrac{1}{2t^2+3}\cdot(4t+0) = \boxed{\dfrac{20t}{2t^2+3}}$

ii. $y = \left(1+2e^{5t}\right)^{1/2} \Rightarrow y' = \frac{1}{2}\left(1+2e^{5t}\right)^{-1/2}\left(0+2\cdot 5e^{5t}\right) = \frac{1}{2}\left(10e^{5t}\right)\left(1+2e^{5t}\right)^{-1/2} = \boxed{\dfrac{5e^{5t}}{\sqrt{1+2e^{5t}}}}$

iii. $y = \left(5x^2+7\right)^{-1/3} \Rightarrow y' = -\frac{1}{3}\left(5x^2+7\right)^{-\frac{1}{3}-1}(10x+0) = -\frac{1}{3}(10x)\left(5x^2+7\right)^{-4/3} = \boxed{\dfrac{-10x}{3\left(5x^2+7\right)^{4/3}}}$

8 $y' = -6x+2(3)e^{3x} = -6x+6e^{3x}$

10 $f'(x) = 30e^{5x}+e^{-x^2}(-2x) = 30e^{5x}-2xe^{-x^2}$

11 $y' = 5e^{5t+1}(5) = 25e^{5t+1}$

12 $w' = e^{-3t^2}(-6t) = -6te^{-3t^2}$

13 $w' = e^{\sqrt{s}} \cdot \frac{1}{2}s^{-1/2} = \dfrac{e^{\sqrt{s}}}{2\sqrt{s}}$

14 $y' = \dfrac{5}{5t+1}$

15 $f'(x) = \dfrac{-1}{1-x} = \dfrac{1}{x-1}$

16 $f'(t) = \dfrac{2t}{t^2+1}$

17 $f'(x) = \dfrac{0-(-e^{-x})}{1-e^{-x}} = \dfrac{e^{-x}}{1-e^{-x}}$

18 $f'(x) = \dfrac{e^x}{e^x+1}$

19 $f'(t) = 5 \cdot \dfrac{5}{5t+1} = \dfrac{25}{5t+1}$

20 $g'(t) = \dfrac{4}{4t+9}$

21 $y' = \dfrac{3}{3t+2}$

23 $y' = 5 + \dfrac{1}{x+2}$

40 $R'(q) = \dfrac{2000q}{1+1000q^2} \Rightarrow R'(10) = \dfrac{20{,}000}{100{,}001} \approx 0.2$
When 10 goods are sold, selling one more increases revenue by about 0.2.

41 $v = \dfrac{ds}{dt} = 20 \cdot \frac{1}{2}e^{t/2} = \boxed{10e^{t/2}}$

42 (a) $\dfrac{dB}{dt} = P\ln\left(1+\dfrac{r}{100}\right)\left(1+\dfrac{r}{100}\right)^t$: how fast the balance is growing, in dollars per year.
(b) $\dfrac{dB}{dr} = Pt\left(1+\dfrac{r}{100}\right)^{t-1} \cdot \dfrac{1}{100} = \dfrac{Pt}{100}\left(1+\dfrac{r}{100}\right)^{t-1}$: about how much more money (in dollars) you would have after t years if the interest rate were 1% higher.

3.3.3 The Chain Rule with Values and Graphs

Ex 4 i. $h'(x) = f'(g(x)) \cdot g'(x)$. At $x = 1$: $h'(1) = f'(g(1)) \cdot g'(1) = f'(0) \cdot (1) = (0)(1) = \boxed{0}$
ii. $k'(x) = g'(f(x)) \cdot f'(x)$. At $x = 1$: $k'(1) = g'(f(1)) \cdot f'(1) = g'(1) \cdot (3) = (1)(3) = \boxed{3}$

Ex 5 To find derivatives, use the slopes of the tangent lines (the lines in the graph).
i. $h'(1) = f'(g(1)) \cdot g'(1)$. From the graph $g(1) = 7$, $g'(1) = \dfrac{7-6}{1-2} = -1$, and $f'(7) = 0$ (horizontal line).
$h'(1) = f'(7) \cdot g'(1) = (0)(-1) = \boxed{0}$
ii. $k'(2) = g'(f(2)) \cdot f'(2)$. From the graph $f(2) = 4$, $f'(2) = \dfrac{4-2}{2-1} = 2$, $g'(4) = \dfrac{3-4}{5-4} = -1$.
$k'(2) = g'(4) \cdot f'(2) = (-1)(2) = \boxed{-2}$

44 From the graphs: f has slope -2 on $0 < x < 40$ and slope 1 on $40 < x < 80$; g has slope $\frac{1}{2}$ on $0 < x < 40$ and slope 0 on $40 < x < 80$.
$\dfrac{d}{dx}f(g(x))\Big|_{x=30} = f'(g(30)) \cdot g'(30) = f'(55) \cdot \frac{1}{2} = (1)\left(\frac{1}{2}\right) = \boxed{\frac{1}{2}}$

45 $f'(g(70)) \cdot g'(70) = f'(60) \cdot 0 = \boxed{0}$

46 $g'(f(30)) \cdot f'(30) = g'(20) \cdot (-2) = \left(\frac{1}{2}\right)(-2) = \boxed{-1}$

47 $g'(f(70)) \cdot f'(70) = g'(30) \cdot (1) = \left(\frac{1}{2}\right)(1) = \boxed{\frac{1}{2}}$

48 $h'(1) = f'(g(1)) \cdot g'(1)$. From the graph $g(1) \approx 2$, $g'(1) \approx 0.8$, $f'(2) \approx 1$.
$h'(1) \approx (1)(0.8) \approx \boxed{0.8}$ (estimate)

49 $k'(1) = g'(f(1)) \cdot f'(1)$. From the graph $f(1) \approx -0.4$, $f'(1) \approx 0.5$, $g'(-0.4) \approx 2$.
$k'(1) \approx (2)(0.5) \approx \boxed{1}$

50 $h'(2) = f'(g(2)) \cdot g'(2)$. From the graph $g(2) \approx 1.6$, $g'(2) \approx -2$, $f'(1.6) \approx 0.7$.
$h'(2) \approx (0.7)(-2) \approx \boxed{-1.4}$ (estimate)

51 $k'(2) = g'(f(2)) \cdot f'(2)$. From the graph $f(2) \approx 0.3$, $f'(2) \approx 1$, $g'(0.3) \approx 1.9$.
$k'(2) \approx (1.9)(1) \approx \boxed{1.9}$

3.3.4 Relative Rates and Logarithms

Ex 6 $\text{RRC} = \dfrac{z'(t)}{z(t)} = \dfrac{P_0 k e^{kt}}{P_0 e^{kt}} \times 100\% = \boxed{k \times 100\%}$

Ex 7 $\text{RRC} = \dfrac{S'}{S} = \dfrac{1095\left(\frac{2}{3}\right) M^{\frac{2}{3}-1}}{1095 M^{2/3}} = \dfrac{\frac{2}{3} M^{-1/3}}{M^{2/3}} = \dfrac{2}{3M^{\frac{2}{3}+\frac{1}{3}}}$

$\boxed{\text{RRC} = \dfrac{2}{3M} \times 100\% \text{ per kg}}$

When $M = 70$ kg: $\text{RRC} = \dfrac{2}{3(70)} \times 100\% \approx 0.95\%$ per kg

The surface area of a human with body mass 70 kilograms increases by about 0.95% if body mass increases by 1 kilogram.

29 $\dfrac{f'}{f} = \dfrac{10}{10t + 5}$

30 $\dfrac{f'}{f} = \dfrac{15}{15t + 12}$

31 $\dfrac{40e^{5t}}{8e^{5t}} = 5$

32 $\dfrac{-210e^{-7t}}{30e^{-7t}} = -7$

33 $\dfrac{12t}{6t^2} = \dfrac{2}{t}$

34 $\dfrac{-140t^{-5}}{35t^{-4}} = -\dfrac{4}{t}$

35 $\ln f = \ln 5 + 1.5t \Rightarrow \dfrac{d}{dt} \ln f = 1.5$

36 $\ln f = \ln 6.8 - 0.5t \Rightarrow \dfrac{d}{dt} \ln f = -0.5$

37 $\ln f = \ln 3 + 2 \ln t \Rightarrow \dfrac{d}{dt} \ln f = \dfrac{2}{t}$

38 $\ln f = \ln 4.5 - 4 \ln t \Rightarrow \dfrac{d}{dt} \ln f = -\dfrac{4}{t}$

43 $f(10) = \sqrt{1001} \approx 31.64$ feet: after 10 seconds the body has traveled about 31.64 feet.
$f'(t) = \frac{1}{2}(t^3+1)^{-1/2}(3t^2) = \dfrac{3t^2}{2\sqrt{t^3+1}} \Rightarrow f'(10) = \dfrac{300}{2\sqrt{1001}} \approx 4.741$ ft/sec: at 10 seconds it is moving at about 4.74 feet per second.
$\dfrac{f'(10)}{f(10)} = \dfrac{4.741}{31.64} \approx 0.15 = 15\%$ per second: the distance is increasing at about 15% per second.

52 (a) $10(1.14)^4 \approx 16.89$ dollars per hour.
(b) With: $16.89e^{0.035(20)}$; without: $10e^{0.035(20)}$. Difference $= (16.89-10)e^{0.7} \approx 6.89(2.014) \approx 13.87$ dollars per hour.
(c) $D(t) = 6.89e^{0.035t}$ is increasing. $D'(t) = 0.035(6.89)e^{0.035t} \Rightarrow D'(20) \approx 0.49$ dollars per hour per year.

53 Tangent at $t = a$ means same value and same slope: $f(a) = Ae^{ka}$ and $f'(a) = h'(a) = kAe^{ka}$.
So $\dfrac{f'(a)}{f(a)} = \dfrac{kAe^{ka}}{Ae^{ka}} = k.$

3.4 The Product and Quotient Rules

In this section

1. The Product Rule 2 lecture · 25 textbook
2. The Quotient Rule 1 lecture · 9 textbook
3. Product and Quotient Rules with Values and Graphs 2 lecture · 5 textbook
4. Tangent Lines and Applications 0 lecture · 8 textbook

3.4.1 The Product Rule

Product rule	$(fg)' = f'g + fg'$ or $\dfrac{d(uv)}{dx} = \dfrac{du}{dx} \cdot v + u \cdot \dfrac{dv}{dx}$
In words	Derivative of the first times the second, plus the first times the derivative of the second.

Lecture Problems

Ex 1 Differentiate

i. $x^2 e^{2x}$

ii. $t^3 \ln(t+1)$

iii. $(3x^2 + 5x)e^x$

Ex 2 Find the derivative of $C = \dfrac{e^{2t}}{t}$.

Textbook Problems

Find the derivative. Assume that a, b, c, and k are constants.

3 $f(x) = xe^x$

4 $f(t) = te^{-2t}$

5 $y = 5xe^{x^2}$

6 $y = t^2(3t + 1)^3$

7 $y = x \ln x$

8 $y = (t^2 + 3)e^t$

9 $z = (3t + 1)(5t + 2)$

10 $y = (t^3 - 7t^2 + 1)e^t$

11 $P = t^2 \ln t$

12 $R = 3qe^{-q}$

15 $y = te^{-t^2}$

16 $f(z) = \sqrt{z}\, e^{-z}$

17 $g(p) = p \ln(2p + 1)$

18 $f(t) = te^{5-2t}$

19 $f(w) = (5w^2 + 3)e^{w^2}$

20 $y = x \cdot 2^x$

21 $w = (t^3 + 5t)(t^2 - 7t + 2)$

22 $z = (te^{3t} + e^{5t})^9$

30 $f(x) = (ax^2 + b)^3$

31 $f(x) = axe^{-bx}$

32 $f(t) = ae^{bt}$

33 $g(\alpha) = e^{\alpha e^{-2\alpha}}$

1 If $f(x) = (2x + 1)(3x - 2)$, find $f'(x)$ two ways: by using the product rule and by multiplying out. Do you get the same result?

2 If $f(x) = x^2(x^3 + 5)$, find $f'(x)$ two ways: by using the product rule and by multiplying out before taking the derivative. Do you get the same result? Should you?

34 If $f(x) = (3x + 8)(2x - 5)$, find $f'(x)$ and $f''(x)$.

3.4.2 The Quotient Rule

Quotient rule $\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$ or $\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\frac{du}{dx}\cdot v - u\cdot\frac{dv}{dx}}{v^2}$

Constant on top If $f(x) = \frac{c}{u(x)}$ (c constant), then $f'(x) = \frac{-c\,u'(x)}{\left(u(x)\right)^2}$

Lecture Problems

Ex 3 Differentiate

i. $\frac{5x^2}{x^3+1}$

ii. $\frac{1}{1+e^x}$

Textbook Problems

Find the derivative. Assume that a, b, c, and k are constants.

13 $f(t) = \frac{5}{t} + \frac{6}{t^2}$

14 $f(x) = \frac{x^2+3}{x}$

23 $f(x) = \frac{x}{e^x}$

24 $w = \frac{3z}{1+2z}$

25 $z = \frac{1-t}{1+t}$

26 $y = \frac{e^x}{1+e^x}$

27 $w = \frac{3y+y^2}{5+y}$

28 $y = \frac{1+z}{\ln z}$

29 $f(x) = \dfrac{ax+b}{cx+k}$

3.4.3 Product and Quotient Rules with Values and Graphs

Product rule $(fg)' = f'g + fg'$ or $\dfrac{d(uv)}{dx} = \dfrac{du}{dx}\cdot v + u\cdot\dfrac{dv}{dx}$

Quotient rule $\left(\dfrac{f}{g}\right)' = \dfrac{f'g - fg'}{g^2}$ or $\dfrac{d}{dx}\left(\dfrac{u}{v}\right) = \dfrac{\frac{du}{dx}\cdot v - u\cdot\frac{dv}{dx}}{v^2}$

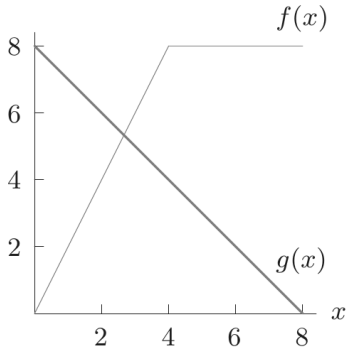
Lecture Problems

Ex 4 If $f(2) = 1$, $f'(2) = 5$, $g(2) = 3$ and $g'(2) = 6$, find

i. $h'(2)$ if $h(x) = f(x)g(x)$

ii. $k'(2)$ if $k(x) = f(x)/g(x)$

Ex 5 Let $h(x) = f(x)g(x)$ and $k(x) = f(x)/g(x)$. Use the given figure to estimate



Figure

i. $h'(2)$

ii. $k'(6)$

Textbook Problems

43 If $\dfrac{d}{dt}\big(tf(t)\big) = 1 + f(t)$, what is $f'(t)$?

44 The quantity, q, of a certain skateboard sold depends on the selling price, p, in dollars, so we write $q = f(p)$. You are given that $f(140) = 15{,}000$ and $f'(140) = -100$.

(a) What do $f(140) = 15{,}000$ and $f'(140) = -100$ tell you about the sales of skateboards?

(b) The total revenue, R, earned by the sale of skateboards is given by $R = pq$. Find $\left.\dfrac{dR}{dp}\right|_{p=140}$.

(c) What is the sign of $\left.\dfrac{dR}{dp}\right|_{p=140}$? If the skateboards are currently selling for 140 dollars, what happens to revenue if the price is increased to 141 dollars?

45 Show that the relative rate of change of a product fg is the sum of the relative rates of change of f and g.

46 Show that the relative rate of change of a quotient f/g is the difference between the relative rates of change of f and g.

47 If $h = f^n$, show that $\dfrac{(f^n)'}{f^n} = n\dfrac{f'}{f}$.

3.4.4 Tangent Lines and Applications

Textbook Problems

35 Find the equation of the tangent line to the graph of $f(x) = x^2 e^{-x}$ at $x = 0$. Check by graphing this function and the tangent line on the same axes.

36 Find the equation of the tangent line to the graph of $f(x) = \dfrac{2x - 5}{x + 1}$ at the point at which $x = 0$.

37 The quantity of a drug, Q mg, present in the body t hours after an injection of the drug is given is $Q = f(t) = 100te^{-0.5t}$. Find $f(1)$, $f'(1)$, $f(5)$, and $f'(5)$. Give units and interpret the answers.

38 A drug concentration curve is given by $C = f(t) = 20te^{-0.04t}$, with C in mg/ml and t in minutes.

(a) Graph C against t. Is $f'(15)$ positive or negative? Is $f'(45)$ positive or negative? Explain.

(b) Find $f(30)$ and $f'(30)$ analytically. Interpret them in terms of the concentration of the drug in the body.

39 For positive constants c and k, the *Monod growth curve* describes the growth of a population, P, as a function of the available quantity of a resource, r: $P = \dfrac{cr}{k + r}$. Find dP/dr and interpret it in terms of the growth of the population.

---

40 If p is price in dollars and q is quantity, demand for a product is given by $q = 5000e^{-0.08p}$.

(a) What quantity is sold at a price of 10 dollars?

(b) Find the derivative of demand with respect to price when the price is 10 dollars and interpret your answer in terms of demand.

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41 The demand for a product is given in Problem 40. Find the revenue and the derivative of revenue with respect to price at a price of 10 dollars. Interpret your answers in economic terms.

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42 The quantity demanded of a certain product, q, is given in terms of p, the price, by $q = 1000e^{-0.02p}$.

(a) Write revenue, R, as a function of price.

(b) Find the rate of change of revenue with respect to price.

(c) Find the revenue and rate of change of revenue with respect to price when the price is 10 dollars. Interpret your answers in economic terms.

Answers · Section 3.4

3.4.1 The Product Rule

Ex 1 i. $y' = (x^2)'(e^{2x}) + (x^2)(e^{2x})' = 2xe^{2x} + x^2(2e^{2x}) = 2xe^{2x} + 2x^2e^{2x} = \boxed{2xe^{2x}(1+x)}$

ii. $w' = (t^3)'\ln(t+1) + t^3\big(\ln(t+1)\big)' = 3t^2\ln(t+1) + t^3 \cdot \dfrac{1}{t+1} = \boxed{3t^2\ln(t+1) + \dfrac{t^3}{t+1}}$

iii. $\big((3x^2+5x)e^x\big)' = (3x^2+5x)'e^x + (3x^2+5x)(e^x)' = \boxed{(6x+5)e^x + (3x^2+5x)e^x}$

Ex 2 **Product rule:** $C = e^{2t}\cdot t^{-1}$

$$C' = (e^{2t})'t^{-1} + e^{2t}(t^{-1})' = 2e^{2t}t^{-1} + e^{2t}(-t^{-2}) = e^{2t}\left(\frac{2t}{t^2} - \frac{1}{t^2}\right) = \boxed{e^{2t}\left(\frac{2t-1}{t^2}\right)}$$

Quotient rule: $C' = \dfrac{(e^{2t})'t - e^{2t}(t)'}{t^2} = \dfrac{2e^{2t}\cdot t - e^{2t}\cdot 1}{t^2} = \boxed{\dfrac{e^{2t}(2t-1)}{t^2}}$

3 $f'(x) = e^x + xe^x = e^x(x+1)$

4 $f'(t) = e^{-2t} + t(-2e^{-2t}) = e^{-2t}(1-2t)$

5 $y' = 5e^{x^2} + 5x(2xe^{x^2}) = 5e^{x^2} + 10x^2e^{x^2}$

6 $y' = 2t(3t+1)^3 + t^2\cdot 3(3t+1)^2(3) = 2t(3t+1)^3 + 9t^2(3t+1)^2$

7 $y' = \ln x + x\cdot\dfrac{1}{x} = \ln x + 1$

8 $y' = 2te^t + (t^2+3)e^t = (t^2+2t+3)e^t$

9 $z' = 3(5t+2) + (3t+1)(5) = 30t + 11$

10 $y' = (3t^2-14t)e^t + (t^3-7t^2+1)e^t = (t^3-4t^2-14t+1)e^t$

11 $P' = 2t\ln t + t^2\cdot\dfrac{1}{t} = 2t\ln t + t$

12 $R' = 3e^{-q} + 3q(-e^{-q}) = 3e^{-q}(1-q)$

15 $y' = e^{-t^2} + t(-2te^{-t^2}) = e^{-t^2}(1-2t^2)$

16 $f'(z) = \frac{1}{2}z^{-1/2}e^{-z} - \sqrt{z}\,e^{-z} = e^{-z}\left(\dfrac{1}{2\sqrt{z}} - \sqrt{z}\right)$

17 $g'(p) = \ln(2p+1) + p\cdot\dfrac{2}{2p+1} = \ln(2p+1) + \dfrac{2p}{2p+1}$

18 $f'(t) = e^{5-2t} + t(-2e^{5-2t}) = e^{5-2t}(1-2t)$

19 $f'(w) = 10we^{w^2} + (5w^2+3)(2we^{w^2}) = 2we^{w^2}(5w^2+8)$

20 $y' = 2^x + x(\ln 2)2^x = 2^x(1 + x\ln 2)$

21 $w' = (3t^2+5)(t^2-7t+2) + (t^3+5t)(2t-7)$

22 $z' = 9(te^{3t}+e^{5t})^8\left(e^{3t}+3te^{3t}+5e^{5t}\right)$

30 $f'(x) = 3(ax^2+b)^2(2ax) = 6ax(ax^2+b)^2$

31 $f'(x) = ae^{-bx} + ax(-be^{-bx}) = ae^{-bx}(1-bx)$

32 $f'(t) = abe^{bt}$

33 $g'(\alpha) = e^{\alpha e^{-2\alpha}}\left(e^{-2\alpha} - 2\alpha e^{-2\alpha}\right) = (1-2\alpha)e^{-2\alpha}e^{\alpha e^{-2\alpha}}$

1 Product rule: $f'(x) = 2(3x-2) + (2x+1)(3) = 6x - 4 + 6x + 3 = 12x - 1$
Multiplying out: $f(x) = 6x^2 - x - 2 \Rightarrow f'(x) = 12x - 1$. Yes, the same result.

2 Product rule: $f'(x) = 2x(x^3+5) + x^2(3x^2) = 2x^4 + 10x + 3x^4 = 5x^4 + 10x$
Multiplying out: $f(x) = x^5 + 5x^2 \Rightarrow f'(x) = 5x^4 + 10x$. Yes, and we should, because it is the same function.

34 $f'(x) = 3(2x-5) + (3x+8)(2) = 12x + 1 \quad f''(x) = 12$

3.4.2 The Quotient Rule

Ex 3 i. $y' = \dfrac{(5x^2)'(x^3+1) - (5x^2)(x^3+1)'}{(x^3+1)^2} = \dfrac{10x(x^3+1) - 5x^2(3x^2)}{(x^3+1)^2} = \dfrac{10x^4 + 10x - 15x^4}{(x^3+1)^2} = \boxed{\dfrac{10x - 5x^4}{(x^3+1)^2}}$

ii. $w' = \dfrac{(1)'(1+e^x) - (1)(1+e^x)'}{(1+e^x)^2} = \dfrac{0(1+e^x) - e^x}{(1+e^x)^2} = \boxed{\dfrac{-e^x}{(1+e^x)^2}}$

13 $f(t) = 5t^{-1} + 6t^{-2} \Rightarrow f'(t) = -\dfrac{5}{t^2} - \dfrac{12}{t^3}$

14 $f'(x) = \dfrac{2x \cdot x - (x^2+3)(1)}{x^2} = \dfrac{x^2 - 3}{x^2}$

23 $f'(x) = \dfrac{e^x - xe^x}{e^{2x}} = \dfrac{1-x}{e^x}$

24 $w' = \dfrac{3(1+2z) - 3z(2)}{(1+2z)^2} = \dfrac{3}{(1+2z)^2}$

25 $z' = \dfrac{(-1)(1+t) - (1-t)(1)}{(1+t)^2} = \dfrac{-2}{(1+t)^2}$

26 $y' = \dfrac{e^x(1+e^x) - e^x \cdot e^x}{(1+e^x)^2} = \dfrac{e^x}{(1+e^x)^2}$

27 $w' = \dfrac{(3+2y)(5+y) - (3y+y^2)(1)}{(5+y)^2} = \dfrac{y^2 + 10y + 15}{(5+y)^2}$

28 $y' = \dfrac{(1)\ln z - (1+z)\frac{1}{z}}{(\ln z)^2} = \dfrac{\ln z - \frac{1}{z} - 1}{(\ln z)^2}$

29 $f'(x) = \dfrac{a(cx+k) - (ax+b)c}{(cx+k)^2} = \dfrac{ak - bc}{(cx+k)^2}$

3.4.3 Product and Quotient Rules with Values and Graphs

Ex 4 i. $h'(x) = f'(x)g(x) + f(x)g'(x)$. At $x = 2$: $h'(2) = f'(2)g(2) + f(2)g'(2) = (5)(3) + (1)(6) = 15 + 6 = \boxed{21}$

ii. $k'(x) = \dfrac{f'(x)g(x) - f(x)g'(x)}{\big(g(x)\big)^2}$. At $x = 2$: $k'(2) = \dfrac{(5)(3) - (1)(6)}{(3)^2} = \dfrac{15-6}{9} = \boxed{1}$

Ex 5 i. From the graph: $f(2) = 4$, $f'(2) = \dfrac{2-4}{1-2} = 2$; $\quad g(2) = 6$, $g'(2) = \dfrac{7-6}{1-2} = -1$.
$h'(2) = f'(2)g(2) + f(2)g'(2) = (2)(6) + (4)(-1) = \boxed{8}$

ii. From the graph: $f(6) = 8$, $f'(6) = 0$ (horizontal line); $\quad g(6) = 2$, $g'(6) = \dfrac{3-2}{5-6} = -1$.
$k'(6) = \dfrac{f'(6)g(6) - f(6)g'(6)}{\big(g(6)\big)^2} = \dfrac{(0)(2) - (8)(-1)}{(2)^2} = \dfrac{8}{4} = \boxed{2}$

43 Product rule: $\dfrac{d}{dt}\big(tf(t)\big) = f(t) + tf'(t)$.

$f(t) + tf'(t) = 1 + f(t) \Rightarrow tf'(t) = 1 \Rightarrow \boxed{f'(t) = \dfrac{1}{t}}$

44 (a) At a price of 140 dollars, 15,000 skateboards are sold; raising the price by 1 dollar decreases sales by about 100 skateboards.

(b) $R = p\,f(p) \Rightarrow \dfrac{dR}{dp} = f(p) + pf'(p) \Rightarrow \left.\dfrac{dR}{dp}\right|_{p=140} = 15{,}000 + 140(-100) = 1000$

(c) Positive. Raising the price to 141 dollars increases revenue by about 1000 dollars.

45 $\dfrac{(fg)'}{fg} = \dfrac{f'g + fg'}{fg} = \dfrac{f'g}{fg} + \dfrac{fg'}{fg} = \boxed{\dfrac{f'}{f} + \dfrac{g'}{g}}$

46 $\dfrac{(f/g)'}{f/g} = \dfrac{f'g - fg'}{g^2} \cdot \dfrac{g}{f} = \dfrac{f'g - fg'}{fg} = \boxed{\dfrac{f'}{f} - \dfrac{g'}{g}}$

47 Chain rule: $(f^n)' = nf^{n-1}f'$, so $\dfrac{(f^n)'}{f^n} = \dfrac{nf^{n-1}f'}{f^n} = \boxed{n\dfrac{f'}{f}}$

3.4.4 Tangent Lines and Applications

35 $y_1 = f(0) = 0 \Rightarrow (0,0)\quad f'(x) = 2xe^{-x} - x^2e^{-x} \Rightarrow m = f'(0) = 0$

$y - 0 = 0(x - 0) \Rightarrow \boxed{y = 0}$ (the x-axis; the graph touches it at its minimum at $x = 0$).

36 $y_1 = f(0) = -5 \Rightarrow (0, -5)$

$f'(x) = \dfrac{2(x+1) - (2x-5)(1)}{(x+1)^2} = \dfrac{7}{(x+1)^2} \Rightarrow m = f'(0) = 7$

$y - (-5) = 7(x - 0) \Rightarrow \boxed{y = 7x - 5}$

37 $f'(t) = 100e^{-0.5t} + 100t(-0.5e^{-0.5t}) = 100e^{-0.5t}(1 - 0.5t)$

$f(1) = 100e^{-0.5} \approx 60.65$ mg: 1 hour after the injection there are about 60.65 mg in the body.

$f'(1) = 50e^{-0.5} \approx 30.33$ mg/hr: at 1 hour the quantity is increasing by about 30.33 mg per hour.

$f(5) = 500e^{-2.5} \approx 41.04$ mg; $\quad f'(5) = 100e^{-2.5}(-1.5) \approx -12.31$ mg/hr: at 5 hours there are about 41.04 mg and it is decreasing by about 12.31 mg per hour.

38 $f'(t) = 20e^{-0.04t} + 20t(-0.04e^{-0.04t}) = 20e^{-0.04t}(1 - 0.04t)$, which is 0 at $t = 25$.

(a) The graph rises to a maximum at $t = 25$ and then falls. So $f'(15) > 0$ (increasing) and $f'(45) < 0$ (decreasing).

(b) $f(30) = 600e^{-1.2} \approx 180.7$ mg/ml: after 30 minutes the concentration is about 180.7 mg/ml.

$f'(30) = 20e^{-1.2}(1 - 1.2) \approx -1.20$ (mg/ml)/min: at 30 minutes the concentration is decreasing by about 1.2 mg/ml per minute.

39 $\dfrac{dP}{dr} = \dfrac{c(k+r) - cr(1)}{(k+r)^2} = \boxed{\dfrac{ck}{(k+r)^2}}$

This is the approximate change in the population per unit increase in the resource. It is always positive (more resource, larger population) and gets smaller as r grows.

40 (a) $q = 5000e^{-0.8} \approx 2247$ items.

(b) $\dfrac{dq}{dp} = 5000(-0.08)e^{-0.08p} = -400e^{-0.08p} \Rightarrow \left.\dfrac{dq}{dp}\right|_{p=10} = -400e^{-0.8} \approx -180$ items per dollar: raising the price from 10 to 11 dollars reduces sales by about 180 items.

41 $R = pq = 5000pe^{-0.08p}$ $\quad R(10) = 50{,}000e^{-0.8} \approx 22{,}466$ dollars.

$R'(p) = 5000e^{-0.08p} + 5000p(-0.08e^{-0.08p}) = 5000e^{-0.08p}(1 - 0.08p)$

$R'(10) = 5000e^{-0.8}(0.2) \approx 449$ dollars per dollar: at a price of 10 dollars the revenue is about 22,466 dollars, and raising the price by 1 dollar increases revenue by about 449 dollars.

42 (a) $R = pq = 1000pe^{-0.02p}$

(b) $R'(p) = 1000e^{-0.02p} + 1000p(-0.02e^{-0.02p}) = 1000e^{-0.02p}(1 - 0.02p)$

(c) $R(10) = 10{,}000e^{-0.2} \approx 8187$ dollars; $\quad R'(10) = 1000e^{-0.2}(0.8) \approx 655$ dollars per dollar. At 10 dollars the revenue is about 8187 dollars, and a 1 dollar price increase raises revenue by about 655 dollars.

3.5 Derivatives of Periodic Functions

In this section

1. Derivatives of Sine and Cosine 2 lecture · 8 textbook
2. The Chain Rule with Sine and Cosine 1 lecture · 15 textbook
3. Applications of Periodic Functions 1 lecture · 7 textbook

3.5.1 Derivatives of Sine and Cosine

Sine and cosine For x in radians: $\dfrac{d}{dx}(\sin x) = \cos x$ and $\dfrac{d}{dx}(\cos x) = -\sin x$

Lecture Problems

Ex 1 Using a calculator, estimate the derivative of $y = \sin x$ at $x = 0$. Make sure your calculator is set in radians.

Ex 2 Differentiate

i. $5\sin t - 8\cos t$

ii. $5 - 3\sin x + x^3$

Textbook Problems

Differentiate. Assume that A and B are constants.

1 $y = 5\sin x$

2 $P = 3 + \cos t$

3 $y = t^2 + 5\cos t$

4 $y = B + A\sin t$

5 $R(q) = q^2 - 2\cos q$

6 $y = 5\sin x - 5x + 4$

21 Find the equation of the tangent line to the graph of $y = \sin x$ at $x = \pi$. Graph the function and the tangent line on the same axes.

24 Find equations of the tangent lines to the graph of $f(x) = \sin x$ at $x = 0$ and at $x = \pi/3$. Use each tangent line to approximate $\sin(\pi/6)$. Would you expect these results to be equally accurate, since they are taken equally far away from $x = \pi/6$ but on opposite sides? If the accuracy is different, can you account for the difference?

3.5.2 The Chain Rule with Sine and Cosine

Chain rule	$\frac{d}{dt}\big(\sin f(t)\big) = f'(t)\cos f(t)$ and $\frac{d}{dt}\big(\cos f(t)\big) = -f'(t)\sin f(t)$
Constant k	$\frac{d}{dt}(\sin kt) = k\cos kt$ and $\frac{d}{dt}(\cos kt) = -k\sin kt$

Lecture Problems

Ex 3 Differentiate

i. $\sin(t^2)$

ii. $5\cos(2t)$

iii. $2x\sin(3x)$

Textbook Problems

Differentiate. Assume that A and B are constants.

7 $f(x) = \sin(3x)$

8 $R = \sin(5t)$

9 $W = 4\cos(t^2)$

10 $y = 2\cos(5t)$

11 $y = \sin(x^2)$

12 $y = A\sin(Bt)$

13 $z = \cos(4\theta)$

14 $y = 6\sin(2t) + \cos(4t)$

15 $f(x) = x^2\cos x$

16 $f(x) = 2x\sin(3x)$

17 $f(\theta) = \theta^3 \cos\theta$

18 $z = \dfrac{e^{t^2} + t}{\sin(2t)}$

19 $f(t) = \dfrac{t^2}{\cos t}$

20 $f(\theta) = \dfrac{\sin\theta}{\theta}$

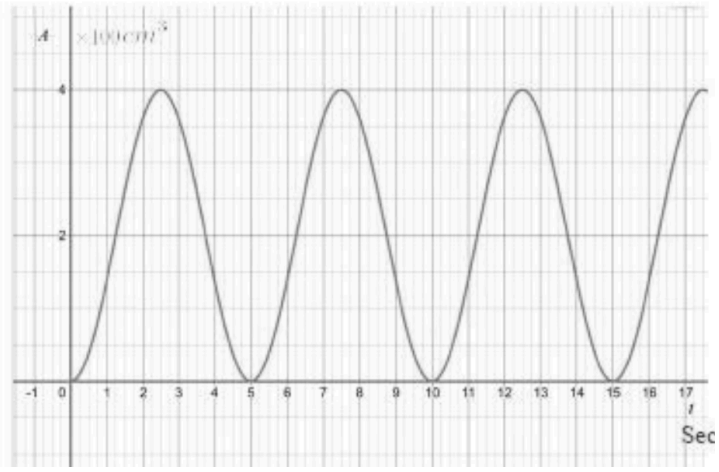
23 Is the graph of $y = \sin(x^4)$ increasing or decreasing when $x = 10$? Is it concave up or concave down?

3.5.3 Applications of Periodic Functions

Period For $\cos(Bt)$ or $\sin(Bt)$: period $P = \dfrac{2\pi}{B}$

Lecture Problems

Ex 4 The average adult takes about 12 breaths per minute. As a patient inhales, the volume of air in the lung increases. As the patient exhales, the volume of air in the lung decreases. For t in seconds since start of the breathing cycle, the volume of air inhaled or exhaled since $t = 0$ is given, in hundreds of cubic centimeters, by $A(t) = -2\cos\left(\dfrac{2\pi}{5}t\right) + 2$.



Volume of air, A (hundreds of cm³), against time t (seconds)

i. How long is one breathing cycle?

ii. Find $A'(1)$ and explain what it means.

Textbook Problems

22 If t is the number of months since June, the number of bird species, N, found in an Ohio forest oscillates approximately according to the formula $N = f(t) = 19 + 9\cos\left(\frac{\pi}{6}t\right)$.

(a) Graph $f(t)$ for $0 \le t \le 24$ and describe what it shows. Use the graph to decide whether $f'(1)$ and $f'(10)$ are positive or negative.

(b) Find $f'(t)$.

(c) Find and interpret $f(1)$, $f'(1)$, $f(10)$, and $f'(10)$.

25 A company's monthly sales, $S(t)$, are seasonal and given as a function of time, t, in months, by $S(t) = 2000 + 600\sin\left(\frac{\pi}{6}t\right)$.

(a) Graph $S(t)$ for $t = 0$ to $t = 12$. What is the maximum monthly sales? What is the minimum monthly sales? If $t = 0$ is January 1, when during the year are sales highest?

(b) Find $S(2)$ and $S'(2)$. Interpret in terms of sales.

26 A boat at anchor is bobbing up and down in the sea. The vertical distance, y, in feet, between the sea floor and the boat is given as a function of time, t, in minutes, by $y = 15 + \sin(2\pi t)$.

(a) Find the vertical velocity, v, of the boat at time t.

(b) Make rough sketches of y and v against t.

27 The depth of the water, y, in meters, in the Bay of Fundy, Canada, is given as a function of time, t, in hours after midnight, by the function $y = 10 + 7.5\cos(0.507t)$. How quickly is the tide rising or falling (in meters/hour) at each of the following times?

(a) 6:00 am

(b) 9:00 am

(c) Noon

(d) 6:00 pm

28 The average adult takes about 12 breaths per minute. As a patient inhales, the volume of air in the lung increases. As the patient exhales, the volume of air in the lung decreases. For t in seconds since start of the breathing cycle, the volume of air inhaled or exhaled since $t = 0$ is given, in hundreds of cubic centimeters, by $A(t) = -2\cos\left(\frac{2\pi}{5}t\right) + 2$.

(a) How long is one breathing cycle?

(b) Find $A'(1)$ and explain what it means.

29 On July 7, 2009, there was a full moon in the Eastern time zone. If t is the number of days since July 7th, the percent of moon illuminated can be represented by $H(t) = 50 + 50\cos\left(\frac{\pi}{15}t\right)$.

(a) Find $H'(t)$. Explain what this tells us about the moon.

(b) For $0 \leq t \leq 30$, when is $H'(t) = 0$? What does this tell us about the moon?

(c) For $0 \leq t \leq 30$, when is $H'(t)$ negative? When is it positive? Explain what positive and negative values of $H'(t)$ tell us about the moon.

30 Paris, France, has a latitude of approximately $49°$ N. If t is the number of days since the start of 2009, the number of hours of daylight in Paris can be approximated by $D(t) = 4\cos\left(\frac{2\pi}{365}(t - 172)\right) + 12$.

(a) Find $D(40)$ and $D'(40)$. Explain what this tells about daylight in Paris.

(b) Find $D(172)$ and $D'(172)$. Explain what this tells about daylight in Paris.

Answers · Section 3.5

3.5.1 Derivatives of Sine and Cosine

Ex 1 Use the average rate of change on the small interval $0 \le x \le 0.01$:

$$f'(0) \approx \frac{\sin(0.01) - \sin(0)}{0.01 - 0} = \frac{0.0099998 - 0}{0.01} = 0.99998 \approx 1.0$$

The derivative of $y = \sin x$ at $x = 0$ is approximately 1.0. *Warning:* x must be in radians.

Ex 2 i. $y' = 5\cos t - 8(-\sin t) = \boxed{5\cos t + 8\sin t}$

ii. $w' = 0 - 3\cos x + 3x^2 = \boxed{-3\cos x + 3x^2}$

1 $y' = 5\cos x$

2 $P' = -\sin t$

3 $y' = 2t - 5\sin t$

4 $y' = A\cos t$

5 $R'(q) = 2q + 2\sin q$

6 $y' = 5\cos x - 5$

21 $y_1 = \sin\pi = 0 \Rightarrow (\pi, 0)$ $\quad y' = \cos x \Rightarrow m = \cos\pi = -1$

$y - 0 = -1(x - \pi) \Rightarrow \boxed{y = -x + \pi}$

24 At $x = 0$: $f(0) = 0$, $m = \cos 0 = 1 \Rightarrow y = x$. Estimate: $\sin(\pi/6) \approx \pi/6 \approx 0.524$.

At $x = \pi/3$: $f = \frac{\sqrt{3}}{2}$, $m = \cos\frac{\pi}{3} = \frac{1}{2} \Rightarrow y = \frac{\sqrt{3}}{2} + \frac{1}{2}\left(x - \frac{\pi}{3}\right)$. Estimate: ≈ 0.604.

True value $\sin(\pi/6) = 0.5$. The estimate from $x = 0$ is more accurate, because near $x = 0$ the sine graph is almost straight (it bends very little), while near $x = \pi/3$ it bends more.

3.5.2 The Chain Rule with Sine and Cosine

Ex 3 i. $y' = \cos(t^2) \cdot (2t) = \boxed{2t\cos(t^2)}$

ii. $w' = 5\big(-\sin(2t) \cdot (2)\big) = \boxed{-10\sin(2t)}$

iii. $f' = (2x)'\sin(3x) + 2x\big(\sin(3x)\big)' = 2\sin(3x) + 2x\big(\cos(3x) \cdot 3\big) =$

$\boxed{2\sin(3x) + 6x\cos(3x)}$

7 $f'(x) = 3\cos(3x)$

8 $R' = 5\cos(5t)$

9 $W' = 4\big(-\sin(t^2)\big)(2t) = -8t\sin(t^2)$

10 $y' = -10\sin(5t)$

11 $y' = 2x\cos(x^2)$

12 $y' = AB\cos(Bt)$

13 $z' = -4\sin(4\theta)$

14 $y' = 12\cos(2t) - 4\sin(4t)$

15 $f'(x) = 2x\cos x - x^2\sin x$

16 $f'(x) = 2\sin(3x) + 6x\cos(3x)$

17 $f'(\theta) = 3\theta^2\cos\theta - \theta^3\sin\theta$

18 $z' = \dfrac{\left(2te^{t^2} + 1\right)\sin(2t) - 2\left(e^{t^2} + t\right)\cos(2t)}{\sin^2(2t)}$

19 $f'(t) = \dfrac{2t\cos t + t^2\sin t}{\cos^2 t}$

20 $f'(\theta) = \dfrac{\theta\cos\theta - \sin\theta}{\theta^2}$

23 $y' = 4x^3\cos(x^4) \Rightarrow y'(10) = 4000\cos(10{,}000) \approx -3809 < 0$: **decreasing**.
$y'' = 12x^2\cos(x^4) - 16x^6\sin(x^4) \Rightarrow y''(10) \approx 4.9 \times 10^6 > 0$: **concave up**.

3.5.3 Applications of Periodic Functions

Ex 4 i. $B = \dfrac{2\pi}{5}$, $P = \dfrac{2\pi}{B} = \boxed{5 \text{ seconds}}$

ii. $A'(t) = -2\left(-\sin\left(\dfrac{2\pi}{5}t\right)\cdot\dfrac{2\pi}{5}\right) + 0 = \dfrac{4\pi}{5}\sin\left(\dfrac{2\pi}{5}t\right)$

$A'(1) = \dfrac{4\pi}{5}\sin\left(\dfrac{2\pi}{5}\right) \approx 2.39$ hundred cm^3/sec

One second after the cycle begins, the patient is inhaling at a rate of about 2.39 hundred cubic centimeters per second, that is 239 cm^3/sec.

22 (a) The number of species oscillates between 10 and 28 with a period of 12 months (highest in June, lowest in December). At $t = 1$ the graph is decreasing, so $f'(1) < 0$; at $t = 10$ it is increasing, so $f'(10) > 0$.

(b) $f'(t) = -9\cdot\dfrac{\pi}{6}\sin\left(\dfrac{\pi}{6}t\right) = -\dfrac{3\pi}{2}\sin\left(\dfrac{\pi}{6}t\right)$

(c) $f(1) \approx 26.8$ species in July; $f'(1) \approx -2.36$ species per month (decreasing). $f(10) = 23.5$ species in April; $f'(10) \approx 4.08$ species per month (increasing).

25 (a) Maximum $2000 + 600 = 2600$; minimum $2000 - 600 = 1400$. Highest when $\sin\left(\frac{\pi}{6}t\right) = 1 \Rightarrow t = 3$: April 1.

(b) $S(2) = 2000 + 600\sin\left(\frac{\pi}{3}\right) \approx 2519.6$: on March 1 monthly sales are about 2520.

$S'(t) = 100\pi\cos\left(\frac{\pi}{6}t\right) \Rightarrow S'(2) = 50\pi \approx 157.1$ per month: sales are increasing by about 157 per month.

26 (a) $v = \dfrac{dy}{dt} = 2\pi\cos(2\pi t)$ feet per minute.

(b) y oscillates between 14 and 16 with period 1 minute; v oscillates between -2π and 2π with the same period, and $v = 0$ at the top and bottom of y.

27 $y' = -7.5(0.507)\sin(0.507t) = -3.8025\sin(0.507t)$

(a) $t = 6$: $y' \approx -0.38$: falling at 0.38 m/hr (b) $t = 9$: $y' \approx 3.76$: rising at 3.76 m/hr

(c) $t = 12$: $y' \approx 0.75$: rising at 0.75 m/hr (d) $t = 18$: $y' \approx -1.12$: falling at 1.12 m/hr

28 Same as lecture Ex 4: (a) 5 seconds (b) $A'(1) \approx 2.39$ hundred cm^3/sec: the patient is inhaling at about 239 cm^3 per second.

29 (a) $H'(t) = -\dfrac{10\pi}{3}\sin\left(\dfrac{\pi}{15}t\right)$: the rate (percent per day) at which the illuminated part of the moon is changing.

(b) $H'(t) = 0$ at $t = 0$, 15, 30 days: full moon ($t = 0$, 30) and new moon ($t = 15$); the percent illuminated is not changing at these moments.

(c) $H'(t) < 0$ for $0 < t < 15$: the illuminated part is shrinking. $H'(t) > 0$ for $15 < t < 30$: it is growing.

30 $D'(t) = -\dfrac{8\pi}{365}\sin\left(\dfrac{2\pi}{365}(t-172)\right)$

(a) $D(40) \approx 9.42$ hours; $D'(40) \approx 0.053$ hours per day: on day 40 (about February 10) there are about 9.4 hours of daylight, increasing by about 3 minutes per day.

(b) $D(172) = 16$ hours; $D'(172) = 0$: on day 172 (about June 21–22) there are 16 hours of daylight (the longest day), and the amount is momentarily not changing.

4.1 Local Maxima and Minima

In this section

1. Local Extrema and Critical Points 1 lecture · 8 textbook
2. The First Derivative Test 3 lecture · 9 textbook
3. The Second Derivative Test 1 lecture · 3 textbook
4. Reading Graphs of f′ and Given Information 2 lecture · 7 textbook
5. Families of Functions and Parameters 1 lecture · 10 textbook

4.1.1 Local Extrema and Critical Points

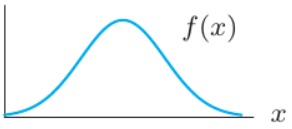
Local minimum	f has a local minimum at p if $f(p)$ is less than or equal to the values of f for points near p.
Local maximum	f has a local maximum at p if $f(p)$ is greater than or equal to the values of f for points near p.
Critical point	A point p in the domain of f where $f'(p) = 0$ or $f'(p)$ is undefined. The point $(p, f(p))$ on the graph is also called a critical point. The **critical value** is $f(p)$.
Where to look	If a continuous function has a local max or min at p, then p is a critical point or an endpoint of the interval. A function may have any number of critical points or none. *Warning:* not every critical point is a local max or min.

Lecture Problems

Ex 1 Find all critical points of $f(x) = x^3 - 9x^2 - 48x + 52$.

Textbook Problems

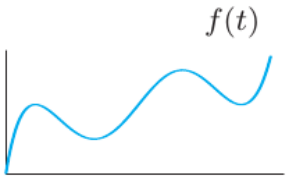
1 In Problems 1–4, indicate all critical points of the function f. How many critical points are there? Identify each critical point as a local maximum, a local minimum, or neither.



Problem 1

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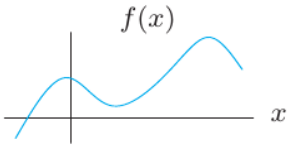
2 In Problems 1–4, indicate all critical points of the function f. How many critical points are there? Identify each critical point as a local maximum, a local minimum, or neither.



Problem 2

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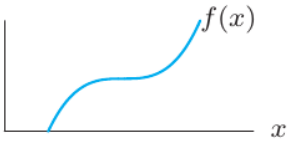
3 In Problems 1–4, indicate all critical points of the function f. How many critical points are there? Identify each critical point as a local maximum, a local minimum, or neither.



Problem 3

---

4 In Problems 1–4, indicate all critical points of the function f. How many critical points are there? Identify each critical point as a local maximum, a local minimum, or neither.



Problem 4

5

(a) Graph a function with two local minima and one local maximum.

(b) Graph a function with two critical points. One of these critical points should be a local minimum, and the other should be neither a local maximum nor a local minimum.

---

6 During an illness a person ran a fever. His temperature rose steadily for eighteen hours, then went steadily down for twenty hours. When was there a critical point for his temperature as a function of time?

---

7 Graph two continuous functions f and g, each of which has exactly five critical points, the points A–E in Figure 4.12, and which satisfy the following conditions:

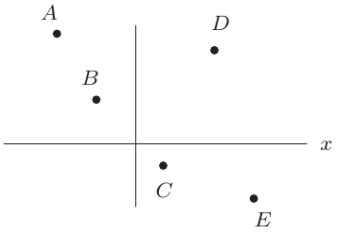


Figure 4.12

(a) $f(x) \to \infty$ as $x \to -\infty$ and $f(x) \to \infty$ as $x \to \infty$

(b) $g(x) \to -\infty$ as $x \to -\infty$ and $g(x) \to 0$ as $x \to \infty$

19 If U and V are positive constants, find all critical points of $F(t) = Ue^{t} + Ve^{-t}$.

4.1.2 The First Derivative Test

First derivative test	If f changes from decreasing to increasing at p (f': $-$ to $+$), f has a **local minimum** at p. If f changes from increasing to decreasing at p (f': $+$ to $-$), f has a **local maximum** at p.
Steps	1. Domain 2. Find f' 3. Solve $f'(x) = 0$ and $f'(x)$ DNE (keep only points in the domain) 4. Sign table with test numbers 5. Conclusion 6. Value $f(p)$

Lecture Problems

Ex 2 Let $f(x) = x^3 - 9x^2 - 48x + 52$. Use the **first derivative test** to confirm that f has a local maximum at $x = -2$ and a local minimum at $x = 8$.

Ex 3 Let $f(x) = \sqrt[3]{x^2 - 4}$.

i. Find the critical points of f.

---

ii. Use the first derivative test to find the local maximum and local minimum values of f.

Ex 7 Find the critical points of the function and classify them as local maxima or local minima or neither.
$g(x) = xe^{-3x}$

Textbook Problems

10 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.
$f(x) = x^3 - 6x + 1$

---

11 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.
$f(x) = x^3 + 6x + 1$

---

12 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.
$f(x) = 3x^5 - 5x^3$

---

13 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.
$f(x) = e^x - 10x$

---

14 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.

$f(x) = x \ln x, \quad x > 0$

---

15 Using a calculator or computer, graph the function. Describe in words the interesting features of the graph, including the location of the critical points and where the function is monotonic (that is, increasing or decreasing). Then use the derivative and algebra to explain the shape of the graph.

$f(x) = x + 2 \sin x$

---

18 The derivative of $f(t)$ is given by $f'(t) = t^3 - 6t^2 + 8t$ for $0 \leq t \leq 5$. Graph $f'(t)$, and describe how the function $f(t)$ changes over the interval $t = 0$ to $t = 5$. When is $f(t)$ increasing and when is it decreasing? Where does $f(t)$ have a local maximum and where does it have a local minimum?

---

23 Find and classify the critical points of $f(x) = x^3(1 - x)^4$ as local maxima and minima.

37

(a) On a computer or calculator, graph $f(\theta) = \theta - \sin\theta$. Can you tell whether the function has any zeros in the interval $0 \le \theta \le 1$?

(b) Find f'. What does the sign of f' tell you about the zeros of f in the interval $0 \le \theta \le 1$?

4.1.3 The Second Derivative Test

Second derivative test	If $f'(p) = 0$ and $f''(p) > 0$ (concave up), f has a **local minimum** at p. If $f'(p) = 0$ and $f''(p) < 0$ (concave down), f has a **local maximum** at p.

Lecture Problems

Ex 4 Let $f(x) = x^3 - 9x^2 - 48x + 52$. Use the **second derivative test** to confirm that f has a local maximum at $x = -2$ and a local minimum at $x = 8$.

Textbook Problems

22 The function $f(x) = x^4 - 4x^3 + 8x$ has a critical point at $x = 1$. Use the second derivative test to identify it as a local maximum or local minimum.

32

(a) If b is a positive constant and $x > 0$, find all critical points of $f(x) = x - b \ln x$.

(b) Use the second derivative test to determine whether the function has a local maximum or local minimum at each critical point.

33

(a) For a a positive constant, find all critical points of $f(x) = x - a\sqrt{x}$.

(b) What value of a gives a critical point at $x = 5$? Does $f(x)$ have a local maximum or a local minimum at this critical point?

4.1.4 Reading Graphs of f′ and Given Information

First derivative test	If f changes from decreasing to increasing at p (f': $-$ to $+$), f has a **local minimum** at p. If f changes from increasing to decreasing at p (f': $+$ to $-$), f has a **local maximum** at p.

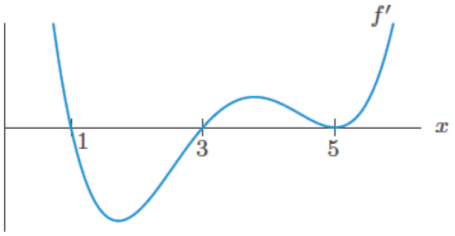
Lecture Problems

Ex 5 Graph a function f with the following properties:

- $f(x)$ has critical points at $x = 2$ and $x = 5$;
- $f'(x)$ is positive to the left of 2 and positive to the right of 5;
- $f'(x)$ is negative between 2 and 5.

Then, identify the critical points as local maxima, local minima, or neither.

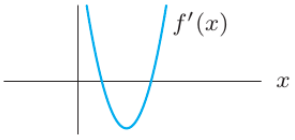
Ex 8 The figure below is a graph of f'. For what values of x does f have a local maximum? A local minimum?



Graph of f′ (not f)

Textbook Problems

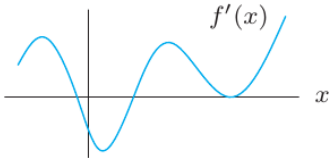
8 Problems 8–9 show the graph of a derivative function f'. Indicate on a sketch the x-values that are critical points of the function f itself. Identify each critical point as a local maximum, a local minimum, or neither.



Problem 8

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9 Problems 8–9 show the graph of a derivative function f'. Indicate on a sketch the x-values that are critical points of the function f itself. Identify each critical point as a local maximum, a local minimum, or neither.



Problem 9

---

16 Figure 4.13 is a graph of f'. For what values of x does f have a local maximum? A local minimum?

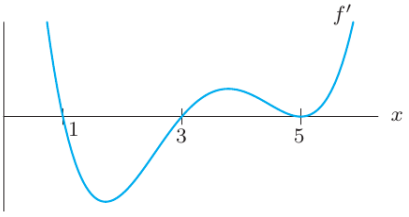


Figure 4.13: Graph of f′ (not f)

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17 On the graph of f' in Figure 4.14, indicate the x-values that are critical points of the function f itself. Are they local maxima, local minima, or neither?

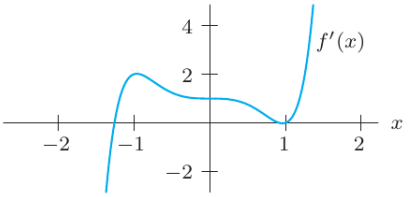


Figure 4.14: Graph of f′ (not f)

20 Consumer demand for a certain product is changing over time, and the rate of change of this demand, $f'(t)$, in units/week, is given, in week t, in the following table.

t	0	1	2	3	4	5	6	7	8	9	10
$f'(t)$	12	10	4	-2	-3	-1	3	7	11	15	10

(a) When is the demand for this product increasing? When is it decreasing?

(b) Approximately when is demand at a local maximum? A local minimum?

21 Suppose f has a continuous derivative whose values are given in the following table.

x	0	1	2	3	4	5	6	7	8	9	10
$f'(x)$	5	2	1	-2	-5	-3	-1	2	3	1	-1

(a) Estimate the x-coordinates of critical points of f for $0 \leq x \leq 10$.

(b) For each critical point, indicate if it is a local maximum of f, local minimum, or neither.

36 Assume f has a derivative everywhere and has just one critical point, at $x = 3$. In parts (a)–(d), you are given additional conditions. In each case decide whether $x = 3$ is a local maximum, a local minimum, or neither. Explain your reasoning. Sketch possible graphs for all four cases.

(a) $f'(1) = 3$ and $f'(5) = -1$

(b) $f(x) \to \infty$ as $x \to \infty$ and as $x \to -\infty$

(c) $f(1) = 1,\ f(2) = 2,\ f(4) = 4,\ f(5) = 5$

(d) $f'(2) = -1,\ f(3) = 1,\ f(x) \to 3$ as $x \to \infty$

4.1.5 Families of Functions and Parameters

Critical point	A point p in the domain of f where $f'(p) = 0$ or $f'(p)$ is undefined. The point $(p, f(p))$ on the graph is also called a critical point. The **critical value** is $f(p)$.
Second derivative test	If $f'(p) = 0$ and $f''(p) > 0$ (concave up), f has a **local minimum** at p. If $f'(p) = 0$ and $f''(p) < 0$ (concave down), f has a **local maximum** at p.

Lecture Problems

Ex 6 Find the critical point of the function $f(x) = x^2 + bx + c$. What is its graphical significance?

Textbook Problems

24 Investigate the one-parameter family of functions. Assume that a is positive.

$f(x) = (x - a)^2$

(a) Graph $f(x)$ using three different values for a.

(b) Using your graph in part (a), describe the critical points of f and how they appear to move as a increases.

(c) Find a formula for the x-coordinates of the critical point(s) of f in terms of a.

25 Investigate the one-parameter family of functions. Assume that a is positive.

$f(x) = x^3 - ax$

(a) Graph $f(x)$ using three different values for a.

(b) Using your graph in part (a), describe the critical points of f and how they appear to move as a increases.

(c) Find a formula for the x-coordinates of the critical point(s) of f in terms of a.

26 Investigate the one-parameter family of functions. Assume that a is positive.
$f(x) = x^2 e^{-ax}$

(a) Graph $f(x)$ using three different values for a.

(b) Using your graph in part (a), describe the critical points of f and how they appear to move as a increases.

(c) Find a formula for the x-coordinates of the critical point(s) of f in terms of a.

27 Find constants a and b so that the minimum for the parabola $f(x) = x^2 + ax + b$ is at the given point. [Hint: Begin by finding the critical point in terms of a.]
$(3, 5)$

28 Find constants a and b so that the minimum for the parabola $f(x) = x^2 + ax + b$ is at the given point. [Hint: Begin by finding the critical point in terms of a.]
$(-2, -3)$

29 Sketch several members of the family $y = x^3 - ax^2$ on the same axes. Discuss the effect of the parameter a on the graph. Find all critical points for this function.

30 For what values of a and b does $f(x) = a(x - b\ln x)$ have a local minimum at the point $(2, 5)$? Figure 4.15 shows a graph of $f(x)$ with $a = 1$ and $b = 1$.

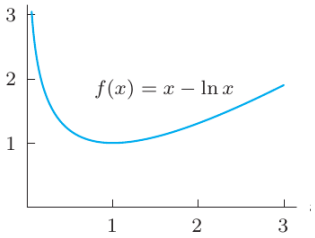


Figure 4.15

31 Find the value of a so that the function $f(x) = xe^{ax}$ has a critical point at $x = 3$.

34 Let $g(x) = x - ke^x$, where k is any constant. For what value(s) of k does the function g have a critical point?

35 If a and b are nonzero constants, find the domain and all critical points of $f(x) = \dfrac{ax^2}{x - b}$.

Answers · Section 4.1

4.1.1 Local Extrema and Critical Points

Ex 1 $D_f = \mathbb{R}$ (polynomial)

$f'(x) = 3x^2 - 18x - 48$

$f'(x) = 0$: $3x^2 - 18x - 48 = 0 \Rightarrow 3(x^2 - 6x - 16) = 0$

$\Rightarrow x^2 - 6x - 16 = 0 \Rightarrow (x+2)(x-8) = 0$

$x = -2 \in D, \quad x = 8 \in D$

(or use the quadratic formula with $a = 3$, $b = -18$, $c = -48$:

$x = \dfrac{18 \pm \sqrt{324 + 576}}{6} = \dfrac{18 \pm 30}{6}$, so $x = 8$ or $x = -2$)

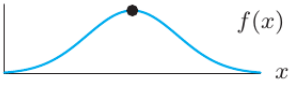
$f'(x)$ DNE: no solution (polynomial).

Critical points: $\boxed{x = -2,\ x = 8}$

Note: the typed notes show $+9x^2$; the doctor corrected it to $-9x^2$, which is the function used in Examples 2 and 4.

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1 **One** critical point: the top of the hump, a **local maximum** (f increases, then decreases).



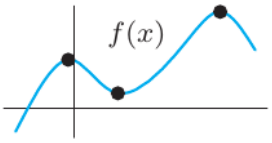
Answer 1

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2 **Four** critical points (where the tangent is horizontal). From left to right: local maximum, local minimum, local maximum, local minimum (after the last one the graph rises again).

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3 **Three** critical points. From left to right: local maximum (near the vertical axis), local minimum, local maximum.

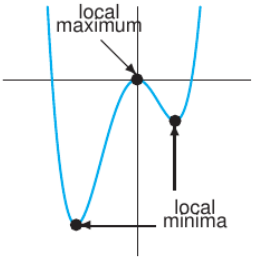


Answer 3

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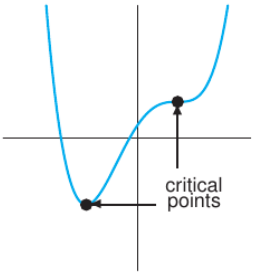
4 **One** critical point: the flat (horizontal) part in the middle. It is **neither** a max nor a min, because f is increasing on both sides.

5 (a) A "W" shape: down to a valley (local min), up to a hilltop (local max), down to a second valley (local min), then up. Example: $f(x) = x^4 - 2x^2$ (min at $x = \pm 1$, max at $x = 0$).



Answer 5(a)

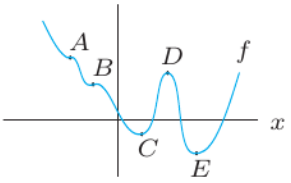
(b) Down to a valley (local min), then up, flatten out for a moment with a horizontal tangent (critical point, neither), then keep going up. Example: $f(x) = x^4 + 4x^3$, $f'(x) = 4x^2(x + 3)$: $x = -3$ local min, $x = 0$ neither.



Answer 5(b)

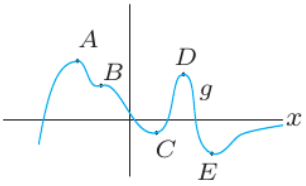
6 At $\boxed{t = 18}$ hours. The temperature changes from increasing to decreasing there, so it is a critical point (a local maximum).

7 (a) Coming down from $+\infty$, f flattens at A and at B (horizontal tangent, neither) while still decreasing, falls to a local min at C, rises to a local max at D, falls to a local min at E, then rises to $+\infty$.



Answer 7(a)

(b) Coming up from $-\infty$, g rises to a local max at A, decreases and flattens at B (neither), falls to a local min at C, rises to a local max at D, falls to a local min at E, then rises toward 0 (approaching the x -axis from below).



Answer 7(b)

19 $F'(t) = Ue^t - Ve^{-t} = 0 \Rightarrow Ue^t = Ve^{-t} \Rightarrow e^{2t} = \dfrac{V}{U} \Rightarrow 2t = \ln\dfrac{V}{U}$

$\boxed{t = 0.5\ln\left(\dfrac{V}{U}\right)}$ (only critical point; F' is never undefined).

Since $F''(t) = Ue^t + Ve^{-t} > 0$, it is a local minimum.

4.1.2 The First Derivative Test

Ex 2 From Example 1: $f'(x) = 3x^2 - 18x - 48$, critical points $x = -2$ and $x = 8$.

Test numbers: $-3, 0, 9$: $f'(-3) = 33 > 0$, $f'(0) = -48 < 0$, $f'(9) = 33 > 0$

x	$-\infty$ to -2	-2 to 8	8 to ∞
$f'(x)$	$+$	$-$	$+$
$f(x)$	↗	↘	↗

f is increasing before -2 and decreasing after -2, so f has a **local max** at $x = -2$:
$f(-2) = (-2)^3 - 9(-2)^2 - 48(-2) + 52 = \boxed{104}$

f is decreasing before 8 and increasing after 8, so f has a **local min** at $x = 8$:
$f(8) = (8)^3 - 9(8)^2 - 48(8) + 52 = \boxed{-396}$

Ex 3 i. $f(x) = (x^2 - 4)^{1/3}$, $D_f = \mathbb{R}$ (cube root is an odd root)

$f'(x) = \frac{1}{3}(x^2 - 4)^{-2/3}(2x) = \dfrac{2x}{3(x^2 - 4)^{2/3}}$

$f'(x) = 0$: $2x = 0 \Rightarrow x = 0 \in D$

$f'(x)$ DNE: $3(x^2 - 4)^{2/3} = 0 \Rightarrow x^2 - 4 = 0 \Rightarrow x = 2,\ x = -2$ (both $\in D$)

Critical points: $\boxed{x = -2,\ 0,\ 2}$

ii. Test numbers $-3, -1, 1, 3$:

x	$-\infty$ to -2	-2 to 0	0 to 2	2 to ∞
$f'(x)$	$-$	$-$	$+$	$+$
$f(x)$	↘	↘	↗	↗

f has a **local min** at $x = 0$: $f(0) = \boxed{\sqrt[3]{-4} \approx -1.587}$

No local max (f' does not change sign at $x = \pm 2$, so they are neither).

Ex 7 $D_g = \mathbb{R}$

$g'(x) = (1)e^{-3x} + x\left(-3e^{-3x}\right) = e^{-3x}(1 - 3x) = \dfrac{1 - 3x}{e^{3x}}$

$g'(x) = 0$: $1 - 3x = 0 \Rightarrow x = \dfrac{1}{3} \in D$

$g'(x)$ DNE: $e^{3x} = 0$, no solution.

Test numbers 0 and 1:

x	$-\infty$ to $\frac{1}{3}$	$\frac{1}{3}$ to ∞
$g'(x)$	$+$	$-$
$g(x)$	↗	↘

g has a **local max** at $x = \dfrac{1}{3}$: $g\left(\dfrac{1}{3}\right) = \dfrac{1}{3}e^{-3(1/3)} = \dfrac{1}{3}e^{-1} = \boxed{\dfrac{1}{3e} \approx 0.123}$

10 Domain: all real numbers $(-\infty, \infty)$.
$f'(x) = 3x^2 - 6 = 0 \Rightarrow x = \pm\sqrt{2}$

x	$-\infty$ to $-\sqrt{2}$	$-\sqrt{2}$ to $\sqrt{2}$	$\sqrt{2}$ to ∞
f'	$+$	$-$	$+$
f	↗	↘	↗

Local max at $x = -\sqrt{2} \approx -1.41$: $f = 1 + 4\sqrt{2} \approx 6.66$.
Local min at $x = \sqrt{2} \approx 1.41$: $f = 1 - 4\sqrt{2} \approx -4.66$.
The graph is a cubic: increasing, then decreasing between $-\sqrt{2}$ and $\sqrt{2}$, then increasing; $f \to -\infty$ as $x \to -\infty$ and $f \to \infty$ as $x \to \infty$.

---

11 Domain: all real numbers $(-\infty, \infty)$.
$f'(x) = 3x^2 + 6 > 0$ for all x ($f' = 0$ has no solution, and f' is defined everywhere).

x	$-\infty$ to ∞
f'	$+$
f	↗

So f is **increasing for all** x and has **no critical points**. The graph rises from $-\infty$ to ∞, crossing the y-axis at 1.

---

12 Domain: all real numbers $(-\infty, \infty)$.
$f'(x) = 15x^4 - 15x^2 = 15x^2(x-1)(x+1) = 0 \Rightarrow x = -1,\ 0,\ 1$

x	$-\infty$ to -1	-1 to 0	0 to 1	1 to ∞
f'	$+$	$-$	$-$	$+$
f	↗	↘	↘	↗

Local max at $x = -1$: $f(-1) = 2$. Local min at $x = 1$: $f(1) = -2$. At $x = 0$ the graph is flat but keeps decreasing: neither.
Increasing for $x < -1$ and $x > 1$, decreasing for $-1 < x < 1$.

---

13 Domain: all real numbers $(-\infty, \infty)$.
$f'(x) = e^x - 10 = 0 \Rightarrow x = \ln 10 \approx 2.3$

x	$-\infty$ to $\ln 10$	$\ln 10$ to ∞
f'	$-$	$+$
f	↘	↗

$f' < 0$ for $x < \ln 10$ (decreasing), $f' > 0$ for $x > \ln 10$ (increasing).
Local min at $(2.3,\ -13.0)$: $f(\ln 10) = 10 - 10\ln 10 \approx -13.03$.
For large negative x the graph looks like the line $y = -10x$; for large positive x it grows like e^x.

---

14 Domain: $x > 0$ (because of $\ln x$).

$f'(x) = \ln x + x \cdot \dfrac{1}{x} = \ln x + 1 = 0 \Rightarrow \ln x = -1 \Rightarrow x = e^{-1} \approx 0.368$

x	0 to $\frac{1}{e}$	$\frac{1}{e}$ to ∞
f'	$-$	$+$
f	↘	↗

$f' < 0$ for $0 < x < \dfrac{1}{e}$ (decreasing), $f' > 0$ for $x > \dfrac{1}{e}$ (increasing).

Local min at $x = \dfrac{1}{e}$: $f\left(\dfrac{1}{e}\right) = -\dfrac{1}{e} \approx -0.368$.

The graph starts near 0 as $x \to 0^+$, dips below the axis to the minimum, crosses the axis at $x = 1$, then increases.

---

15 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 1 + 2\cos x = 0 \Rightarrow \cos x = -\frac{1}{2} \Rightarrow x = \dfrac{2\pi}{3} + 2k\pi$ or $x = \dfrac{4\pi}{3} + 2k\pi$

On one period $0 \le x \le 2\pi$ (the pattern repeats every 2π):

x	0 to $\frac{2\pi}{3}$	$\frac{2\pi}{3}$ to $\frac{4\pi}{3}$	$\frac{4\pi}{3}$ to 2π
f'	$+$	$-$	$+$
f	↗	↘	↗

$f' < 0$ when $\cos x < -\frac{1}{2}$ (between $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$, plus multiples of 2π), $f' > 0$ otherwise.

Local maxima at $x = \dfrac{2\pi}{3} + 2k\pi$, local minima at $x = \dfrac{4\pi}{3} + 2k\pi$.

The function is **alternately increasing and decreasing**; the graph wiggles around the line $y = x$ and climbs overall.

---

18 $f'(t) = t(t^2 - 6t + 8) = t(t-2)(t-4)$: zero at $t = 0, 2, 4$. The graph of f' is a cubic starting at 0, positive (a hump) on $0 < t < 2$, negative on $2 < t < 4$, positive again after 4 ($f'(5) = 15$).

t	0 to 2	2 to 4	4 to 5
f'	$+$	$-$	$+$
f	↗	↘	↗

f is increasing for $0 < t < 2$ and $4 < t < 5$, decreasing for $2 < t < 4$.

Local max at $t = 2$, local min at $t = 4$.

23 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 3x^2(1-x)^4 + x^3 \cdot 4(1-x)^3(-1) = x^2(1-x)^3\big[3(1-x) - 4x\big] = x^2(1-x)^3(3-7x)$

$f'(x) = 0 \Rightarrow x = 0,\ x = \dfrac{3}{7},\ x = 1$

x	$-\infty$ to 0	0 to $\frac{3}{7}$	$\frac{3}{7}$ to 1	1 to ∞
f'	$+$	$+$	$-$	$+$
f	↗	↗	↘	↗

$x = 0$: **not a max or min**. $x = \dfrac{3}{7}$: **local max** ($f \approx 0.0084$). $x = 1$: **local min** ($f(1) = 0$).

37 (a) $f(0) = 0$, so $\theta = 0$ is a zero. Near 0 the graph is so flat (very close to the axis) that you cannot tell from the graph whether there are other zeros in $0 \le \theta \le 1$.

(b) $f'(\theta) = 1 - \cos\theta > 0$ for $0 < \theta \le 1$. So f is increasing on this interval: starting at $f(0) = 0$ it only goes up. **The only zero is at the origin**, $\theta = 0$.

4.1.3 The Second Derivative Test

Ex 4 From Example 1: $f'(x) = 3x^2 - 18x - 48$, critical points $x = -2$ and $x = 8$.

$f''(x) = 6x - 18$

$x = -2$: $f''(-2) = 6(-2) - 18 = -30 < 0$, so by the 2nd derivative test f has a **local max** at $x = -2$.

$x = 8$: $f''(8) = 6(8) - 18 = 30 > 0$, so by the 2nd derivative test f has a **local min** at $x = 8$.

22 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 4x^3 - 12x^2 + 8$, $f'(1) = 4 - 12 + 8 = 0$ (critical point).

$f''(x) = 12x^2 - 24x$, $f''(1) = 12 - 24 = -12 < 0$ (concave down).

By the 2nd derivative test, $x = 1$ is a **local maximum** ($f(1) = 5$).

32 (a) $f'(x) = 1 - \dfrac{b}{x} = 0 \Rightarrow \boxed{x = b}$ (the only critical point; f' is defined for $x > 0$).

(b) $f''(x) = \dfrac{b}{x^2}$, $f''(b) = \dfrac{1}{b} > 0$, so f has a **local minimum** at $x = b$.

33 (a) Domain $x \ge 0$. $f'(x) = 1 - \dfrac{a}{2\sqrt{x}}$.

$f'(x) = 0 \Rightarrow \sqrt{x} = \dfrac{a}{2} \Rightarrow x = \dfrac{a^2}{4}$; $f'(x)$ DNE at $x = 0$ (in the domain). Critical points:

$\boxed{x = 0,\ x = \dfrac{a^2}{4}}$

(b) $\dfrac{a^2}{4} = 5 \Rightarrow a^2 = 20 \Rightarrow \boxed{a = \sqrt{20} = 2\sqrt{5} \approx 4.47}$

$f''(x) = \dfrac{a}{4}x^{-3/2} > 0$ at $x = 5$, so it is a **local minimum**.

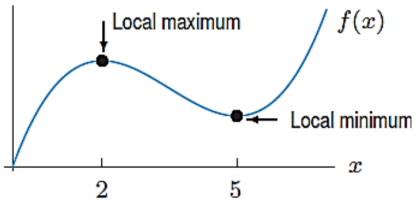
4.1.4 Reading Graphs of f′ and Given Information

Ex 5

x	$-\infty$ to 2	2 to 5	5 to ∞
$f'(x)$	$+$	$-$	$+$
$f(x)$	↗	↘	↗

f has a **local max at** $x = 2$ (concave down there, $f''(2) < 0$) and a **local min at** $x = 5$ (concave up there, $f''(5) > 0$).

Graph: rising until $x = 2$ (a hilltop), falling from $x = 2$ to $x = 5$, then rising again after $x = 5$ (a valley at $x = 5$). One possible graph:



A function with critical points at x = 2 and x = 5

Ex 8 $f'(1) = 0$, $f'(3) = 0$, $f'(5) = 0$, so $x = 1$, $x = 3$ and $x = 5$ are critical points of f.

x	$-\infty$ to 1	1 to 3	3 to 5	5 to ∞
$f'(x)$	$+$	$-$	$+$	$+$
$f(x)$	↗	↘	↗	↗

f has a **local max at** $x = 1$ and a **local min at** $x = 3$. At $x = 5$, f' does not change sign, so it is neither.

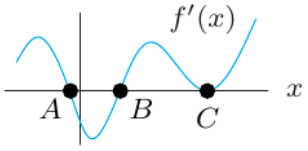
8 f' crosses the x-axis twice, so f has two critical points.

First zero (just to the right of the vertical axis): f' goes from $+$ to $-$, so **local maximum**.

Second zero: f' goes from $-$ to $+$, so **local minimum**.

9 Three critical points (zeros of f'), call them A, B, C from left to right:

A: f' goes $+$ to $-$: **local max**. B: f' goes $-$ to $+$: **local min**. C: f' touches 0 but stays positive: **neither**.



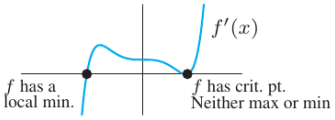
Answer 9

16 Critical points: $f' = 0$ at $x = 1, 3, 5$.

x	left of 1	1 to 3	3 to 5	right of 5
f'	$+$	$-$	$+$	$+$
f	↗	↘	↗	↗

Local max at $x = 1$, **local min at** $x = 3$; $x = 5$ is neither.

17 $f' = 0$ at $x \approx -1.3$ and at $x = 1$.
At $x \approx -1.3$: f' goes from $-$ to $+$, so f has a **local min**.
At $x = 1$: f' touches 0 but is positive on both sides, so it is a critical point that is **neither** a max nor a min.



Answer 17

20 (a) Demand is increasing when $f' > 0$: for $0 < t < 2.5$ and $5.5 < t < 10$ (approximately). Decreasing when $f' < 0$: for $2.5 < t < 5.5$.
(b) f' changes from $+$ to $-$ between $t = 2$ and $t = 3$: **local max at about** $t = 2.5$ weeks. f' changes from $-$ to $+$ between $t = 5$ and $t = 6$: **local min at about** $t = 5.5$ weeks.

21 (a) f' changes sign (so $f' = 0$) between 2 and 3, between 6 and 7, and between 9 and 10: $x \approx 2.5$, $x \approx 6.5$, $x \approx 9.5$.
(b) $x \approx 2.5$: f' goes $+$ to $-$: **local max**. $x \approx 6.5$: $-$ to $+$: **local min**. $x \approx 9.5$: $+$ to $-$: **local max**.

36 Since $x = 3$ is the only critical point, f' cannot change sign except at $x = 3$: f is increasing or decreasing on all of $x < 3$, and on all of $x > 3$.
(a) $f'(1) > 0$: increasing left of 3; $f'(5) < 0$: decreasing right of 3. **Local maximum.** Sketch: a single hill with its top at $x = 3$.
(b) f must come down from ∞ and go back up to ∞: decreasing then increasing. **Local minimum.** Sketch: a single valley (U shape) with its bottom at $x = 3$.
(c) $f(1) < f(2)$: increasing left of 3; $f(4) < f(5)$: increasing right of 3. **Neither.** Sketch: always rising, with a flat spot (horizontal tangent) at $x = 3$.
(d) $f'(2) < 0$: decreasing left of 3. Right of 3, f goes from $f(3) = 1$ toward 3, so it increases. **Local minimum.** Sketch: falling to a low point $(3, 1)$, then rising and leveling off toward the line $y = 3$.

4.1.5 Families of Functions and Parameters

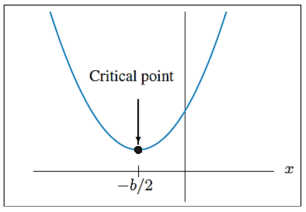
Ex 6 $D_f = \mathbb{R}$ (polynomial)

$f'(x) = 2x + b$

$f'(x) = 0 \Rightarrow 2x + b = 0 \Rightarrow 2x = -b \Rightarrow \boxed{x = -\frac{b}{2}} \in D$

$f''(x) = 2, \quad f''\left(-\frac{b}{2}\right) = 2 > 0$, so by the 2nd derivative test f has a **local min** at $x = -\frac{b}{2}$.

Graphical significance: the critical point $x = -\frac{b}{2}$ is the **vertex of the parabola**, the graph of the quadratic function.



Critical point of the parabola

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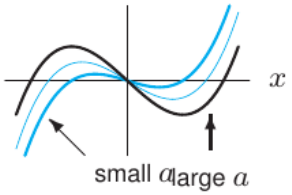
24 (a) For $a = 1, 2, 3$: three identical upward parabolas with vertices at $(1, 0)$, $(2, 0)$, $(3, 0)$.

(b) Each graph has one critical point, a local minimum at the vertex on the x-axis. As a increases, it moves to the right.

(c) $f'(x) = 2(x - a) = 0 \Rightarrow \boxed{x = a}$

---

25 (a) For example $a = 1, 2, 3$: cubics through the origin, each with a hump (local max) to the left of the origin and a dip (local min) to the right; larger a gives a bigger hump and dip.



Answer 25(a)

(b) Two critical points: a local max at negative x and a local min at positive x. As a increases they **move farther from the origin**.

(c) $f'(x) = 3x^2 - a = 0 \Rightarrow \boxed{x = \pm\sqrt{a/3}}$ ($-\sqrt{a/3}$: local max, $+\sqrt{a/3}$: local min)

---

26 (a) For $a = 1, 2, 3$: each graph touches the origin (minimum), rises to a hump, then decays toward 0 as $x \to \infty$ (and grows to ∞ as $x \to -\infty$). Larger a gives a lower hump closer to the origin.

(b) Two critical points: a local min at $x = 0$ (does not move) and a local max at $x > 0$, which moves toward the origin (and gets lower: $f(2/a) = \frac{4}{a^2 e^2}$) as a increases.

(c) $f'(x) = 2xe^{-ax} - ax^2 e^{-ax} = xe^{-ax}(2 - ax) = 0 \Rightarrow \boxed{x = 0,\ x = \frac{2}{a}}$

---

27 $f'(x) = 2x + a = 0 \Rightarrow x = -\frac{a}{2}$. We need $-\frac{a}{2} = 3 \Rightarrow a = -6$.

$f(3) = 9 - 18 + b = 5 \Rightarrow b = 14$. $\boxed{a = -6,\ b = 14}$

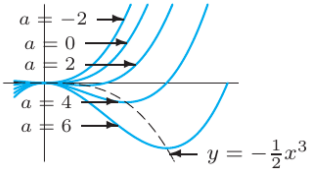
28 $f'(x) = 2x + a = 0 \Rightarrow x = -\dfrac{a}{2}$. We need $-\dfrac{a}{2} = -2 \Rightarrow a = 4$.

$f(-2) = 4 - 8 + b = -3 \Rightarrow b = 1$. $\boxed{a = 4,\ b = 1}$

29 $y' = 3x^2 - 2ax = x(3x - 2a) = 0 \Rightarrow \boxed{x = 0,\ x = \dfrac{2a}{3}}$

For $a > 0$: local max at $x = 0$ (value 0), local min at $x = \dfrac{2a}{3}$ (value $-\dfrac{4a^3}{27}$). For $a < 0$ the roles swap (max at $\frac{2a}{3}$, min at 0). For $a = 0$, $y = x^3$ has one critical point, $x = 0$, which is neither.

Effect of a: every graph passes through the origin; as a (positive) increases, the dip moves right and gets deeper. The local minima lie on the curve $y = -\frac{1}{2}x^3$.



Answer 29

30 $f'(x) = a\left(1 - \dfrac{b}{x}\right) = 0 \Rightarrow x = b$. Need $x = 2$, so $b = 2$.

$f(2) = a(2 - 2\ln 2) = 5 \Rightarrow a = \dfrac{5}{2 - 2\ln 2} \approx 8.15$

Check: $f''(x) = \dfrac{ab}{x^2} > 0$, so it is a local minimum. $\boxed{a = \dfrac{5}{2 - 2\ln 2} \approx 8.15,\ b = 2}$

31 $f'(x) = e^{ax} + axe^{ax} = e^{ax}(1 + ax) = 0 \Rightarrow 1 + ax = 0 \Rightarrow x = -\dfrac{1}{a}$

$-\dfrac{1}{a} = 3 \Rightarrow \boxed{a = -\dfrac{1}{3}}$

34 $g'(x) = 1 - ke^x = 0 \Rightarrow e^x = \dfrac{1}{k}$.

This has a solution only if $\dfrac{1}{k} > 0$. So g has a critical point when $\boxed{k > 0}$, at $x = \ln\dfrac{1}{k} = -\ln k$. For $k \leq 0$ there is no critical point.

35 Domain: all real numbers except $x = b$.

$f'(x) = \dfrac{2ax(x - b) - ax^2(1)}{(x - b)^2} = \dfrac{ax^2 - 2abx}{(x - b)^2} = \dfrac{ax(x - 2b)}{(x - b)^2}$

$f'(x) = 0 \Rightarrow x = 0,\ x = 2b$. f' is undefined at $x = b$, but b is not in the domain.

Critical points: $\boxed{x = 0,\ x = 2b}$

4.2 Inflection Points

In this section

1. Finding Inflection Points with the Second Derivative 3 lecture · 2 textbook
2. Graphing a Function from Its Properties 1 lecture · 8 textbook
3. Inflection Points in Applications 1 lecture · 8 textbook
4. Critical Points and Inflection Points Together 1 lecture · 13 textbook
5. Inflection Points from a Graph of f, f′ or f″ 3 lecture · 5 textbook

4.2.1 Finding Inflection Points with the Second Derivative

Inflection point	A point at which the graph of a function f **changes concavity** is called an inflection point of f.
Concavity	$f'' > 0$: concave up (C.U.) $\quad$ $f'' < 0$: concave down (C.D.)
Possible inflection point	If f'' is zero or undefined at p ($f''(p) = 0$ or $f''(p)$ DNE), then p is a possible inflection point.
Test	p is an inflection point only if f'' **changes sign** at p.
Steps	Domain $\to f' \to f'' \to$ solve $f'' = 0$ and f'' DNE $\to$ sign table of $f'' \to$ inflection point $\big(p, f(p)\big)$.
Warning	Not every point where $f''(x) = 0$ (or f'' is undefined) is an inflection point (e.g. $f(x) = x^4$ at $x = 0$), just as not every point where $f'(x) = 0$ is a local maximum or minimum.

Lecture Problems

Ex 1 Find the inflection points of $f(x) = x^3 - 9x^2 - 48x + 52$.

Ex 5 Let $f(x) = x^4$. Show that the graph of f has no inflection points.

Ex 6 What is the concavity of the graph of $f(x) = ax^2 + bx + c$?

Textbook Problems

10 For $f(x) = x^3 - 18x^2 - 10x + 6$, find the inflection point algebraically. Graph the function with a calculator or computer and confirm your answer.

21 Find the inflection points of $f(x) = x^4 + x^3 - 3x^2 + 2$.

4.2.2 Graphing a Function from Its Properties

Concavity $f'' > 0$: concave up (C.U.) $f'' < 0$: concave down (C.D.)

Lecture Problems

Ex 2 Graph a function f with the following properties: f has a critical point at $x = 4$ and an inflection point at $x = 8$; the value of f' is negative to the left of 4 and positive to the right of 4; the value of f'' is positive to the left of 8 and negative to the right of 8.

Textbook Problems

5

(a) Graph a polynomial with two local maxima and two local minima.

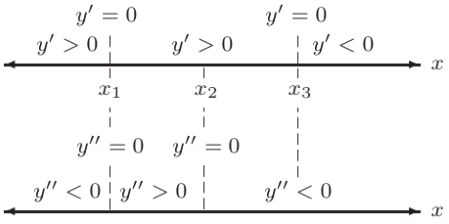
(b) What is the least number of inflection points this function must have? Label the inflection points.

6 Graph a function with only one critical point (at $x = 5$) and one inflection point (at $x = 10$). Label the critical point and the inflection point on your graph.

7 Graph a function which has a critical point and an inflection point at the same place.

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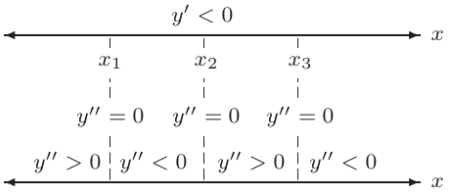
23 Sketch a possible graph of $y = f(x)$, using the given information about the derivatives $y' = f'(x)$ and $y'' = f''(x)$. Assume that the function is defined and continuous for all real x.



Problem 23

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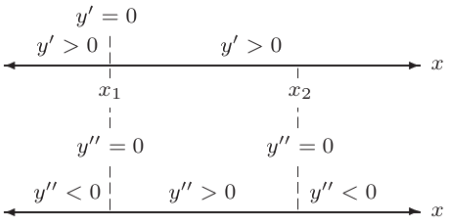
24 Sketch a possible graph of $y = f(x)$, using the given information about the derivatives $y' = f'(x)$ and $y'' = f''(x)$. Assume that the function is defined and continuous for all real x.



Problem 24

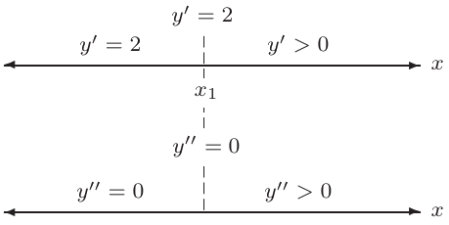
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25 Sketch a possible graph of $y = f(x)$, using the given information about the derivatives $y' = f'(x)$ and $y'' = f''(x)$. Assume that the function is defined and continuous for all real x.



Problem 25

26 Sketch a possible graph of $y = f(x)$, using the given information about the derivatives $y' = f'(x)$ and $y'' = f''(x)$. Assume that the function is defined and continuous for all real x.



Problem 26

36 Assume that the polynomial f has exactly two local maxima and one local minimum, and that these are the only critical points of f.

(a) Sketch a possible graph of f.

(b) What is the largest number of zeros f could have?

(c) What is the least number of zeros f could have?

(d) What is the least number of inflection points f could have?

(e) What is the smallest degree f could have?

(f) Find a possible formula for $f(x)$.

4.2.3 Inflection Points in Applications

Meaning	Increasing and C.U.: increasing faster and faster. Increasing and C.D.: increasing slower and slower. At the inflection point the quantity is changing fastest (or slowest).

Lecture Problems

Ex 3 Figure 4.20 shows a population growing toward a limiting population, L. There is an inflection point on the graph at the point where the population reaches $L/2$. What is the significance of the inflection point to the population?

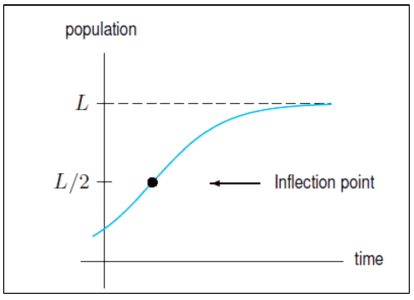


Figure 4.20

Textbook Problems

8 During a flood, the water level in a river first rose faster and faster, then rose more and more slowly until it reached its highest point, then went back down to its pre-flood level. Consider water depth as a function of time.

(a) Is the time of highest water level a critical point or an inflection point of this function?

(b) Is the time when the water first began to rise more slowly a critical point or an inflection point?

9 When I got up in the morning I put on only a light jacket because, although the temperature was dropping, it seemed that the temperature would not go much lower. But I was wrong. Around noon a northerly wind blew up and the temperature began to drop faster and faster. The worst was around 6 pm when, fortunately, the temperature started going back up.

(a) When was there a critical point in the graph of temperature as a function of time?

(b) When was there an inflection point in the graph of temperature as a function of time?

27 In 1774, Captain James Cook left 10 rabbits on a small Pacific island. The rabbit population is approximated by $P(t) = \dfrac{2000}{1 + e^{5.3 - 0.4t}}$ with t measured in years since 1774. Using a calculator or computer:

(a) Graph P. Does the population level off?

(b) Estimate when the rabbit population grew most rapidly. How large was the population at that time?

(c) Find the inflection point on the graph and explain its significance for the rabbit population.

(d) What natural causes could lead to the shape of the graph of P?

28

(a) Water is flowing at a constant rate (i.e., constant volume per unit time) into a cylindrical container standing vertically. Sketch a graph showing the depth of water against time.

(b) Water is flowing at a constant rate into a cone-shaped container standing on its point. Sketch a graph showing the depth of the water against time.

29 If water is flowing at a constant rate (i.e., constant volume per unit time) into the Grecian urn in Figure 4.25, sketch a graph of the depth of the water against time. Mark on the graph the time at which the water reaches the widest point of the urn.

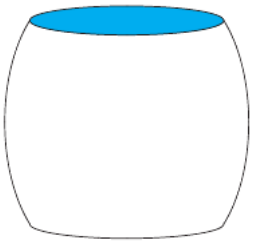


Figure 4.25

30 If water is flowing at a constant rate (i.e., constant volume per unit time) into the vase in Figure 4.26, sketch a graph of the depth of the water against time. Mark on the graph the time at which the water reaches the corner of the vase.

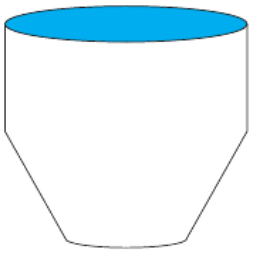


Figure 4.26

31 Water flows at a constant rate into the left side of the W-shaped container in Figure 4.27. Sketch a graph of the height, H, of the water in the left side of the container as a function of time, t. The container starts empty.



Figure 4.27

32 The vase in Figure 4.28 is filled with water at a constant rate (i.e., constant volume per unit time).

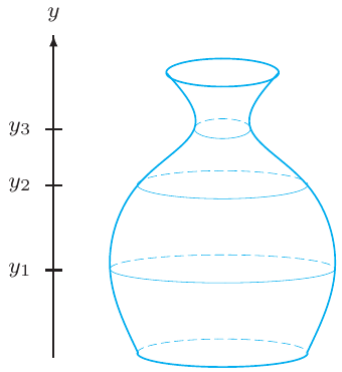


Figure 4.28

(a) Graph $y = f(t)$, the depth of the water, against time, t. Show on your graph the points at which the concavity changes.

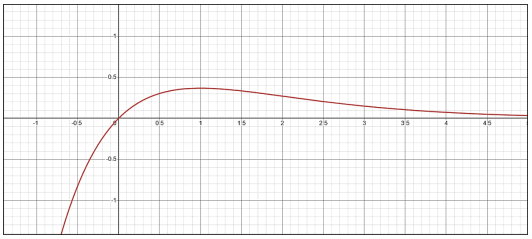
(b) At what depth is $y = f(t)$ growing most quickly? Most slowly? Estimate the ratio between the growth rates at these two depths.

4.2.4 Critical Points and Inflection Points Together

Steps Domain $\to f' \to f'' \to$ solve $f'' = 0$ and f'' DNE $\to$ sign table of $f'' \to$ inflection point $\big(p, f(p)\big)$.

Lecture Problems

Ex 4



Graph of f(x) = xe^{-x}

i. How many critical points and how many inflection points does the function $f(x) = xe^{-x}$ have?

ii. Use derivatives to find the critical points and inflection points exactly

Textbook Problems

11 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = x^2 - 5x + 3$

12 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.

$f(x) = x^3 - 3x + 10$

---

13 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.

$f(x) = 2x^3 + 3x^2 - 36x + 5$

---

14 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.

$f(x) = \frac{x^3}{6} + \frac{x^2}{4} - x + 2$

---

15 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.

$f(x) = x^4 - 2x^2$

---

16 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = 3x^4 - 4x^3 + 6$

17 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = x^4 - 8x^2 + 5$

18 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = x^4 - 4x^3 + 10$

19 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = x^5 - 5x^4 + 35$

20 Use the first derivative to find all critical points and use the second derivative to find all inflection points. Use a graph to identify each critical point as a local maximum, a local minimum, or neither.
$f(x) = 3x^5 - 5x^3$

22

(a) Find all critical points and all inflection points of the function $f(x) = x^4 - 2ax^2 + b$. Assume a and b are positive constants.

(b) Find values of the parameters a and b if f has a critical point at the point $(2, 5)$.

(c) If there is a critical point at $(2, 5)$, where are the inflection points?

---

33 Find a formula for the function described: A cubic polynomial, $ax^3 + bx^2 + cx + d$, with a critical point at $x = 2$, an inflection point at $(1, 4)$, and a leading coefficient of 1.

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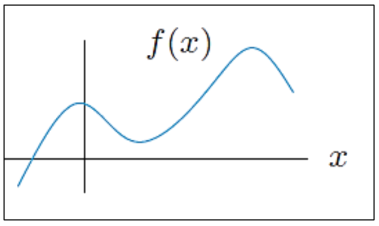
34 Find a formula for the function described: A function of the form $y = bxe^{-ax}$ with a local maximum at $(3, 6)$.

4.2.5 Inflection Points from a Graph of f, f′ or f″

From the graph of f′	Inflection points of f are where f' changes from increasing to decreasing or vice versa (local max or min of f').
From the graph of f″	Inflection points of f are where f'' changes sign (the graph of f'' crosses the x-axis, not just touches it).

Lecture Problems

Ex 7 Indicate on the graph of the function f in the figure below the x-values that are inflection points of the function f.



Ex 8 Indicate on the graph of the function f' in the figure below the x-values that are inflection points of the function f.

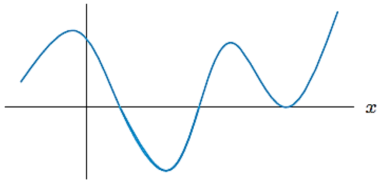


Figure of f'

Ex 9 Indicate on the graph of the function f'' in the figure below the x-values that are inflection points of the function f.

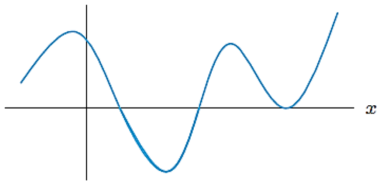
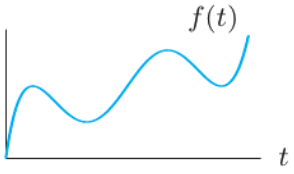


Figure of f''

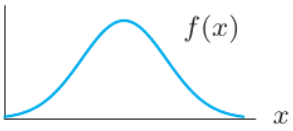
Textbook Problems

1 Indicate the approximate locations of all inflection points. How many inflection points are there?



Problem 1

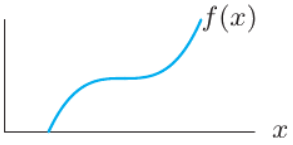
2 Indicate the approximate locations of all inflection points. How many inflection points are there?



Problem 2

---

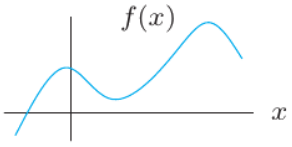
3 Indicate the approximate locations of all inflection points. How many inflection points are there?



Problem 3

---

4 Indicate the approximate locations of all inflection points. How many inflection points are there?



Problem 4

---

35 Indicate on Figure 4.29 approximately where the inflection points of $f(x)$ are if the graph shows

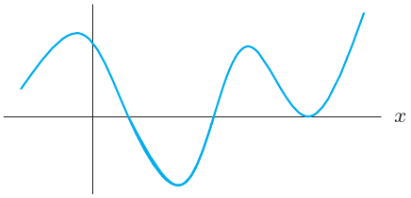


Figure 4.29

(a) The function $f(x)$

(b) The derivative $f'(x)$

(c) The second derivative $f''(x)$

Answers · Section 4.2

4.2.1 Finding Inflection Points with the Second Derivative

Ex 1 $D_f = \mathbb{R}$ (polynomial)

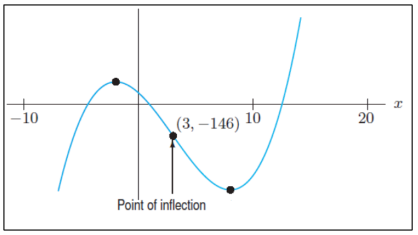
$f'(x) = 3x^2 - 18x - 48 \quad f''(x) = 6x - 18$

$f''(x) = 0 \Rightarrow 6x - 18 = 0 \Rightarrow x = 3 \in D_f \quad f''(x)$ DNE: no solution.

x	$(-\infty, 3)$	$(3, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

f has an inflection point at $x = 3$. $f(3) = (3)^3 - 9(3)^2 - 48(3) + 52 = -146$

Inflection point: $\boxed{(3, -146)}$



Graph of f(x)

Ex 5 $D_f = \mathbb{R}$ (polynomial) $\quad f'(x) = 4x^3 \quad f''(x) = 12x^2$

$f'' = 0 \Rightarrow 12x^2 = 0 \Rightarrow x^2 = 0 \Rightarrow x = 0 \in D_f \quad f''$ DNE: no solution.

x	$(-\infty, 0)$	$(0, \infty)$
f''	$+$	$+$
f	C.U.	C.U.

f'' does not change sign at $x = 0$, so f has **no** inflection point at $x = 0$ (and so none at all).

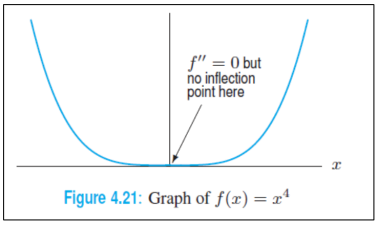


Figure 4.21: Graph of $f(x) = x^4$

Ex 6 $D_f = \mathbb{R}$ (polynomial) $\quad f'(x) = 2ax + b \quad f''(x) = 2a$

$f'' = 0$: no solution; $\quad f''$ DNE: no solution. So f has no inflection point.

f is concave up on $\mathbb{R}$ if $a > 0$; $\quad f$ is concave down on $\mathbb{R}$ if $a < 0$.

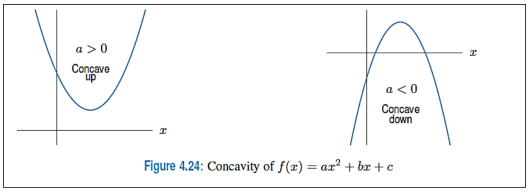


Figure 4.24: Concavity of $f(x) = ax^2 + bx + c$

10 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 3x^2 - 36x - 10 \quad f''(x) = 6x - 36$

$f''(x) = 0 \Rightarrow 6x = 36 \Rightarrow x = 6$

x	$(-\infty, 6)$	$(6, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

$f(6) = 216 - 648 - 60 + 6 = -486$. Inflection point: $\boxed{(6, -486)}$. The graph changes from concave down to concave up there.

21 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 4x^3 + 3x^2 - 6x \quad f''(x) = 12x^2 + 6x - 6 = 6(2x - 1)(x + 1)$

$f''(x) = 0 \Rightarrow x = -1,\ \frac{1}{2}$

x	$(-\infty, -1)$	$(-1, \frac{1}{2})$	$(\frac{1}{2}, \infty)$
f''	$+$	$-$	$+$
f	C.U.	C.D.	C.U.

Inflection points at $x = -1$ and $x = \frac{1}{2}$: $(-1, -1)$ and $\left(\frac{1}{2}, \frac{23}{16}\right)$.

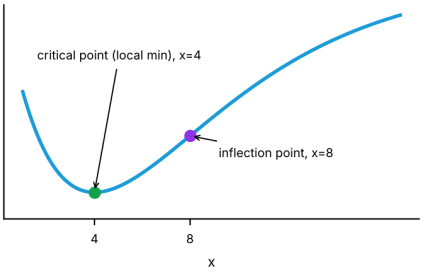
4.2.2 Graphing a Function from Its Properties

Ex 2

x	$(-\infty, 4)$	$(4, 8)$	$(8, \infty)$
f'	$-$	$+$	$+$
f''	$+$	$+$	$-$
f	↘ C.U.	↗ C.U.	↗ C.D.

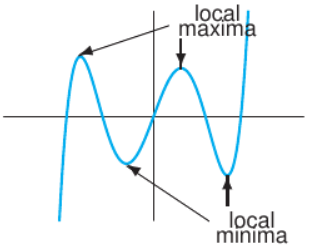
Local minimum at $x = 4$; inflection point at $x = 8$.

The graph: decreasing and concave up until $x = 4$ (lowest point), then increasing and concave up (getting steeper) until $x = 8$, then still increasing but concave down (flattening out) after $x = 8$.

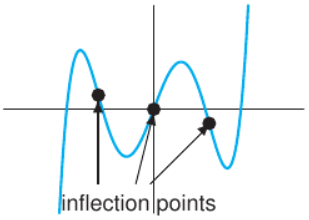


One possible graph

5 (a) For example: up to a local max, down to a local min, up to a local max, down to a local min, then up forever (a polynomial of degree 5 with positive leading coefficient).



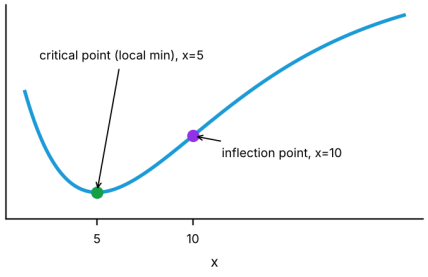
(b) **3**: between each local max and the next local min (or min and next max) the concavity must change, so there is an inflection point between each pair of neighbouring extrema: 4 extrema give at least 3 inflection points.



6 For example:

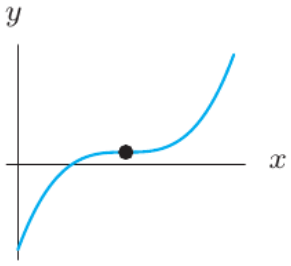
x	$(-\infty, 5)$	$(5, 10)$	$(10, \infty)$
f'	$-$	$+$	$+$
f''	$+$	$+$	$-$
f	↘ C.U.	↗ C.U.	↗ C.D.

Decreasing and C.U. down to a local min at $x = 5$ (the critical point), then increasing and getting steeper until $x = 10$ (the inflection point, where it is steepest), then still increasing but flattening out (C.D.). (Same idea as lecture Ex 2.)



One possible graph

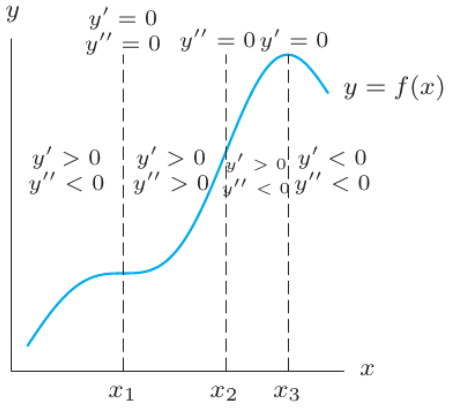
7 For example $f(x) = x^3$ at $x = 0$: $f'(x) = 3x^2$, so $f'(0) = 0$ (critical point); $f''(x) = 6x$ changes from $-$ to $+$ at $x = 0$ (inflection point). The graph is increasing, C.D. before the point, flat at the point, C.U. after it.



23

x	$< x_1$	(x_1, x_2)	(x_2, x_3)	$> x_3$
y'	$+$	$+$	$+$	$-$
y''	$-$	$+$	$-$	$-$
f	↗ C.D.	↗ C.U.	↗ C.D.	↘ C.D.

Increasing and C.D. up to x_1, where the graph is flat (horizontal tangent) and changes to C.U. (inflection point, not a max/min); increasing and C.U. to x_2 (inflection point, steepest point); increasing and C.D. up to a local max at x_3; then decreasing and C.D.



24

x	$< x_1$	(x_1, x_2)	(x_2, x_3)	$> x_3$
y'	$-$	$-$	$-$	$-$
y''	$+$	$-$	$+$	$-$
f	↘ C.U.	↘ C.D.	↘ C.U.	↘ C.D.

The graph is decreasing everywhere (no critical points). It is C.U. before x_1, C.D. on (x_1, x_2), C.U. on (x_2, x_3) and C.D. after x_3: three inflection points, at x_1, x_2, x_3 (a falling "wavy" curve).

25

x	$< x_1$	(x_1, x_2)	$> x_2$
y'	$+$	$+$	$+$
y''	$-$	$+$	$-$
f	↗ C.D.	↗ C.U.	↗ C.D.

The graph is increasing everywhere. C.D. up to x_1, where it is momentarily flat ($y' = 0$: a critical point that is neither max nor min, and an inflection point); then C.U. (getting steeper) up to x_2 (inflection point, steepest point); then C.D. (flattening) after x_2.

26 For $x < x_1$: $y' = 2$ and $y'' = 0$, so the graph is a **straight line with slope 2**.
For $x > x_1$: $y' > 0$ and $y'' > 0$, so the graph is increasing and C.U.: it bends upward, getting steeper than slope 2.
So: a straight line of slope 2 up to x_1, which then curves smoothly upward (concave up) after x_1. (x_1 is not an inflection point, since y'' does not change sign there.)

36 (a) An "M" shape: rising to a local max, falling to a local min, rising to a second local max, then falling; both ends go down to $-\infty$.
(b) **4** (if the local min is below the x-axis and both maxima are above it).
(c) **0** (if both local maxima are below the x-axis).
(d) **2** (C.D. near each max, C.U. near the min).
(e) **4** (f' must have 3 zeros, so f' has degree at least 3).
(f) For example $f(x) = 2x^2 - x^4$: $f'(x) = 4x - 4x^3 = 4x(1-x)(1+x)$, local maxima at $x = \pm 1$, local min at $x = 0$.

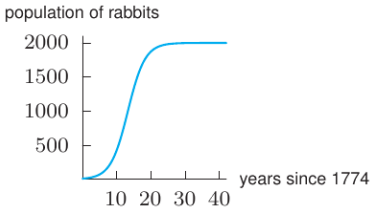
4.2.3 Inflection Points in Applications

Ex 3 • At times before the inflection point, the population is increasing faster every year (increasing and C.U.: accelerating).
• At times after the inflection point the population is increasing slower every year (increasing and C.D.: decelerating).
• At the inflection point, the population is growing fastest.

8 (a) A **critical point** (a local maximum: the depth stops rising and starts falling, so the derivative is 0).
(b) An **inflection point** (the depth changes from rising faster and faster, C.U., to rising more and more slowly, C.D.).

9 (a) Around **6 pm**: the temperature stopped falling and started rising (a local minimum).
(b) Around **noon**: the temperature changed from falling more slowly (C.U.) to falling faster and faster (C.D.). There is another inflection point between noon and 6 pm, where the fall started to slow down again (C.D. to C.U.) before the minimum at 6 pm.

27 (a) The graph is an S-shaped curve rising from about 10 toward 2000. Yes, it levels off at **2000 rabbits** (as $t \to \infty$, $e^{5.3-0.4t} \to 0$, so $P \to 2000$).



(b) The population grows fastest at the inflection point, where $e^{5.3-0.4t} = 1 \Rightarrow 5.3 - 0.4t = 0$ $\Rightarrow t = 13.25$, i.e. around **1787**. Then $P = \dfrac{2000}{1+1} = 1000$ rabbits.

(c) Inflection point $\approx (13.25,\ 1000)$: in 1787, with 1000 rabbits (half the limiting value). Before it the population grew faster and faster (C.U.); after it, it grew more and more slowly (C.D.).

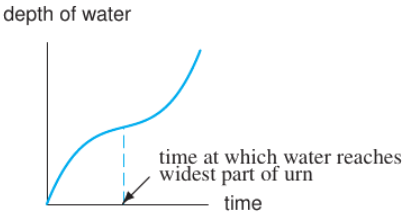
(d) At first there is plenty of food and space, so the rabbits multiply quickly; later limited food and space (and predators, disease) slow the growth, so the population levels off at what the island can support.

28 (a) The cylinder has the same width at every height, so the depth rises at a constant rate: the graph is a **straight line** through the origin (increasing, no concavity).

(b) The cone gets wider as the water rises, so the depth rises quickly at first and then more and more slowly: the graph is **increasing and concave down** (very steep at the start).

29 As the urn gets wider (bottom to middle), the depth rises more and more slowly: increasing and C.D. As it gets narrower (middle to top), the depth rises faster and faster: increasing and C.U.

The time at which the water reaches the widest point is the **inflection point** (where the depth is rising most slowly).

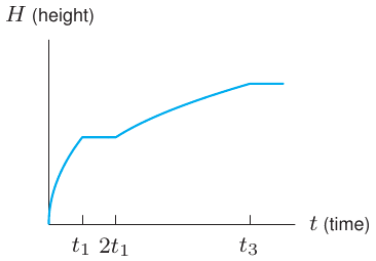


30 Bottom part: the vase gets wider as it goes up, so the depth rises more and more slowly: increasing and **concave down**.

Top part (straight sides, constant width): the depth rises at a constant rate: a **straight line**.

So the graph is a concave-down curve that becomes a straight line at the time the water reaches the corner (mark that time on the t-axis).

31 • $0 < t < t_1$: the left side gets wider as it rises, so H increases more and more slowly (C.D.), until the water reaches the top of the middle divider at t_1.
• $t_1 < t < 2t_1$: the water spills into the right side, so H stays **constant** (horizontal line) while the right side (same size) fills.
• $2t_1 < t < t_3$: both sides fill together over a wide, widening region, so H increases slowly and C.D.
• $t > t_3$: the container is full, H is constant.



32 (a) Bottom to y_1: the vase gets wider, so the depth rises more and more slowly (C.D.). From y_1 to y_3: the vase gets narrower, so the depth rises faster and faster (C.U.). Above y_3: wider again, so C.D. The graph is always increasing, with concavity changes (inflection points) at depths y_1 (widest point) and y_3 (narrowest point, the neck).
(b) Most quickly at y_3 (narrowest); most slowly at y_1 (widest). The growth rate is inversely proportional to the cross-sectional area, i.e. to (diameter)^2. From the figure the diameter at y_1 is about 4 times the diameter at y_3, so the depth grows about $4^2 = 16$ times faster at y_3 than at y_1 (rough estimate from the picture).

4.2.4 Critical Points and Inflection Points Together

Ex 4 i. From the graph: **one critical point** (at $x = 1$, f has a local max: the graph goes up then down) and **one inflection point** (at $x = 2$).
ii. $D_f = \mathbb{R}$ because $D_x = \mathbb{R}$ and $D_{e^{-x}} = \mathbb{R}$.
$f'(x) = (1)e^{-x} + x(-e^{-x}) = e^{-x}(1 - x) = \dfrac{1 - x}{e^x}$
$f'(x) = 0 \Rightarrow 1 - x = 0 \Rightarrow x = 1 \in D_f$ $\quad f'(x)$ DNE: $e^x = 0$, no solution. $\quad x = 1$ is a critical point.

x	$(-\infty, 1)$	$(1, \infty)$
f'	$+$	$-$
f	↗	↘

At $x = 1$, f has a local max: $f(1) = e^{-1} \approx 0.37$.
$f''(x) = (-e^{-x})(1 - x) + e^{-x}(-1) = -e^{-x}(1 - x + 1) = \dfrac{-(2 - x)}{e^x}$
$f''(x) = 0 \Rightarrow 2 - x = 0 \Rightarrow x = 2 \in D_f$ $\quad f''(x)$ DNE: $e^x = 0$, no solution.

x	$(-\infty, 2)$	$(2, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

f has an inflection point at $x = 2$: $\left(2, 2e^{-2}\right) \approx (2, 0.27)$.

11 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 2x - 5 = 0 \Rightarrow x = \frac{5}{2}$

x	$(-\infty, \frac{5}{2})$	$(\frac{5}{2}, \infty)$
f'	$-$	$+$
f	↘	↗

Critical point $x = \frac{5}{2}$: **local minimum** ($f(\frac{5}{2}) = -3.25$).

$f''(x) = 2 > 0$ always: concave up everywhere, **no inflection points**.

---

12 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 3x^2 - 3 = 3(x-1)(x+1) = 0 \Rightarrow x = -1,\ 1$

x	$(-\infty, -1)$	$(-1, 1)$	$(1, \infty)$
f'	$+$	$-$	$+$
f	↗	↘	↗

$x = -1$: **local max** ($f(-1) = 12$); $x = 1$: **local min** ($f(1) = 8$).

$f''(x) = 6x = 0 \Rightarrow x = 0$

x	$(-\infty, 0)$	$(0, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

Inflection point at $x = 0$: $(0, 10)$.

---

13 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 6x^2 + 6x - 36 = 6(x+3)(x-2) = 0 \Rightarrow x = -3,\ 2$

x	$(-\infty, -3)$	$(-3, 2)$	$(2, \infty)$
f'	$+$	$-$	$+$
f	↗	↘	↗

$x = -3$: **local max** ($f(-3) = 86$); $x = 2$: **local min** ($f(2) = -39$).

$f''(x) = 12x + 6 = 0 \Rightarrow x = -\frac{1}{2}$

x	$(-\infty, -\frac{1}{2})$	$(-\frac{1}{2}, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

Inflection point at $x = -\frac{1}{2}$: $(-\frac{1}{2},\ 23.5)$.

14 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = \dfrac{x^2}{2} + \dfrac{x}{2} - 1 = \dfrac{(x+2)(x-1)}{2} = 0 \Rightarrow x = -2,\ 1$

x	$(-\infty, -2)$	$(-2, 1)$	$(1, \infty)$
f'	$+$	$-$	$+$
f	↗	↘	↗

$x = -2$: **local max** ($f(-2) = \frac{11}{3} \approx 3.67$); $x = 1$: **local min** ($f(1) = \frac{17}{12} \approx 1.42$).

$f''(x) = x + \frac{1}{2} = 0 \Rightarrow x = -\frac{1}{2}$

x	$(-\infty, -\frac{1}{2})$	$(-\frac{1}{2}, \infty)$
f''	$-$	$+$
f	C.D.	C.U.

Inflection point at $x = -\frac{1}{2}$: $\left(-\frac{1}{2}, \frac{61}{24}\right) \approx (-0.5,\ 2.54)$.

---

15 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 4x^3 - 4x = 4x(x-1)(x+1) = 0 \Rightarrow x = -1,\ 0,\ 1$

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
f'	$-$	$+$	$-$	$+$
f	↘	↗	↘	↗

$x = -1$: **local min** ($f = -1$); $x = 0$: **local max** ($f = 0$); $x = 1$: **local min** ($f = -1$).

$f''(x) = 12x^2 - 4 = 0 \Rightarrow x^2 = \frac{1}{3} \Rightarrow x = \pm\dfrac{1}{\sqrt{3}}$

x	$(-\infty, -\frac{1}{\sqrt{3}})$	$(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}})$	$(\frac{1}{\sqrt{3}}, \infty)$
f''	$+$	$-$	$+$
f	C.U.	C.D.	C.U.

Inflection points at $x = -\dfrac{1}{\sqrt{3}}$ and $x = \dfrac{1}{\sqrt{3}}$ (both with $f = -\frac{5}{9}$).

---

16 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 12x^3 - 12x^2 = 12x^2(x-1) = 0 \Rightarrow x = 0,\ 1$

x	$(-\infty, 0)$	$(0, 1)$	$(1, \infty)$
f'	$-$	$-$	$+$
f	↘	↘	↗

$x = 0$: **neither** (f' does not change sign); $x = 1$: **local min** ($f(1) = 5$).

$f''(x) = 36x^2 - 24x = 12x(3x-2) = 0 \Rightarrow x = 0,\ \frac{2}{3}$

x	$(-\infty, 0)$	$(0, \frac{2}{3})$	$(\frac{2}{3}, \infty)$
f''	$+$	$-$	$+$
f	C.U.	C.D.	C.U.

Inflection points at $x = 0$: $(0, 6)$ and $x = \frac{2}{3}$: $\left(\frac{2}{3}, \frac{146}{27}\right) \approx (0.67,\ 5.41)$.

17 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 4x^3 - 16x = 4x(x-2)(x+2) = 0 \Rightarrow x = -2,\ 0,\ 2$

x	$(-\infty, -2)$	$(-2, 0)$	$(0, 2)$	$(2, \infty)$
f'	$-$	$+$	$-$	$+$
f	↘	↗	↘	↗

$x = 0$: **local max** ($f = 5$); $x = \pm 2$: **local minima** ($f = -11$).

$f''(x) = 12x^2 - 16 = 0 \Rightarrow x^2 = \frac{4}{3} \Rightarrow x = \pm\dfrac{2}{\sqrt{3}}$

x	$(-\infty, -\frac{2}{\sqrt{3}})$	$(-\frac{2}{\sqrt{3}}, \frac{2}{\sqrt{3}})$	$(\frac{2}{\sqrt{3}}, \infty)$
f''	$+$	$-$	$+$
f	C.U.	C.D.	C.U.

Inflection points at $x = \pm\dfrac{2}{\sqrt{3}} \approx \pm 1.15$ (both with $f = -\frac{35}{9}$).

18 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 4x^3 - 12x^2 = 4x^2(x-3) = 0 \Rightarrow x = 0,\ 3$

x	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
f'	$-$	$-$	$+$
f	↘	↘	↗

$x = 0$: **neither**; $x = 3$: **local min** ($f(3) = -17$).

$f''(x) = 12x^2 - 24x = 12x(x-2) = 0 \Rightarrow x = 0,\ 2$

x	$(-\infty, 0)$	$(0, 2)$	$(2, \infty)$
f''	$+$	$-$	$+$
f	C.U.	C.D.	C.U.

Inflection points at $x = 0$: $(0, 10)$ and $x = 2$: $(2, -6)$.

19 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 5x^4 - 20x^3 = 5x^3(x-4) = 0 \Rightarrow x = 0,\ 4$

x	$(-\infty, 0)$	$(0, 4)$	$(4, \infty)$
f'	$+$	$-$	$+$
f	↗	↘	↗

$x = 0$: **local max** ($f(0) = 35$); $x = 4$: **local min** ($f(4) = -221$).

$f''(x) = 20x^3 - 60x^2 = 20x^2(x-3) = 0 \Rightarrow x = 0,\ 3$

x	$(-\infty, 0)$	$(0, 3)$	$(3, \infty)$
f''	$-$	$-$	$+$
f	C.D.	C.D.	C.U.

$x = 0$ is not an inflection point (f'' does not change sign). Inflection point at $x = 3$: $(3, -127)$.

20 Domain: all real numbers $(-\infty, \infty)$.

$f'(x) = 15x^4 - 15x^2 = 15x^2(x-1)(x+1) = 0 \Rightarrow x = -1,\ 0,\ 1$

x	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
f'	$+$	$-$	$-$	$+$
f	↗	↘	↘	↗

$x = -1$: **local max** ($f = 2$); $x = 0$: **neither**; $x = 1$: **local min** ($f = -2$).

$f''(x) = 60x^3 - 30x = 30x(2x^2 - 1) = 0 \Rightarrow x = 0,\ \pm\dfrac{1}{\sqrt{2}}$

x	$(-\infty, -\frac{1}{\sqrt{2}})$	$(-\frac{1}{\sqrt{2}}, 0)$	$(0, \frac{1}{\sqrt{2}})$	$(\frac{1}{\sqrt{2}}, \infty)$
f''	$-$	$+$	$-$	$+$
f	C.D.	C.U.	C.D.	C.U.

Inflection points at $x = -\dfrac{1}{\sqrt{2}}$, $x = 0$, $x = \dfrac{1}{\sqrt{2}}$: $\left(-\frac{1}{\sqrt{2}}, \frac{7\sqrt{2}}{8}\right)$, $(0, 0)$, $\left(\frac{1}{\sqrt{2}}, -\frac{7\sqrt{2}}{8}\right)$ ($\frac{7\sqrt{2}}{8} \approx 1.24$).

---

22 (a) $f'(x) = 4x^3 - 4ax = 4x(x - \sqrt{a})(x + \sqrt{a}) = 0 \Rightarrow x = 0,\ \pm\sqrt{a}$

x	$(-\infty, -\sqrt{a})$	$(-\sqrt{a}, 0)$	$(0, \sqrt{a})$	$(\sqrt{a}, \infty)$
f'	$-$	$+$	$-$	$+$
f	↘	↗	↘	↗

Critical points: $x = 0$ (local max, $f = b$) and $x = \pm\sqrt{a}$ (local minima, $f = b - a^2$).

$f''(x) = 12x^2 - 4a = 0 \Rightarrow x = \pm\sqrt{\dfrac{a}{3}}$; f'' is $+, -, +$, so both are inflection points.

(b) Critical point at $x = 2$: $\sqrt{a} = 2 \Rightarrow a = 4$. Then $f(2) = 16 - 32 + b = 5 \Rightarrow b = 21$. So $\boxed{a = 4,\ b = 21}$.

(c) $x = \pm\sqrt{\dfrac{4}{3}} = \pm\dfrac{2}{\sqrt{3}} \approx \pm 1.15$ (with $f = \frac{109}{9} \approx 12.1$).

---

33 $a = 1$: $f(x) = x^3 + bx^2 + cx + d$, $f'(x) = 3x^2 + 2bx + c$, $f''(x) = 6x + 2b$

Inflection at $x = 1$: $f''(1) = 6 + 2b = 0 \Rightarrow b = -3$

Critical point at $x = 2$: $f'(2) = 12 - 12 + c = 0 \Rightarrow c = 0$

$f(1) = 4$: $1 - 3 + 0 + d = 4 \Rightarrow d = 6$

$\boxed{y = x^3 - 3x^2 + 6}$

---

34 $y' = be^{-ax} + bx(-ae^{-ax}) = be^{-ax}(1 - ax) = 0 \Rightarrow x = \dfrac{1}{a}$

Local max at $x = 3$: $\dfrac{1}{a} = 3 \Rightarrow a = \dfrac{1}{3}$

$y(3) = 6$: $3be^{-1} = 6 \Rightarrow b = 2e \approx 5.44$ ($b > 0$, so y' goes from $+$ to $-$: a maximum)

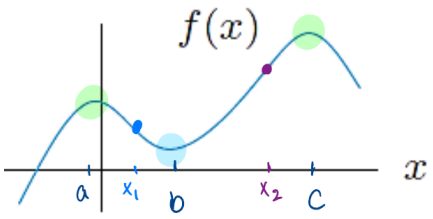
$\boxed{y = 2e\, xe^{-x/3}}$

4.2.5 Inflection Points from a Graph of f, f′ or f″

Ex 7 Let a = the first local max, b = the local min, c = the second local max of the graph.
Inflection points at x_1 (between a and b, where the curve changes from C.D. to C.U.) and x_2 (between b and c, where it changes from C.U. to C.D.).

x	$(-\infty, x_1)$	(x_1, x_2)	(x_2, ∞)
f''	$-$	$+$	$-$
f	C.D.	C.U.	C.D.

f has a local max at $x = a$ and $x = c$, and a local min at $x = b$.

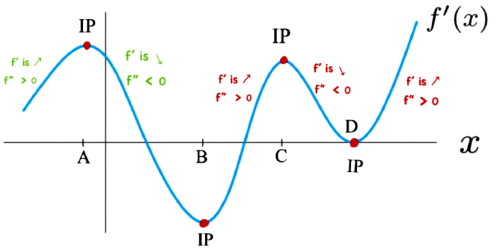


The doctor's labelled graph

Ex 8 Inflection points of f are where f' changes from increasing to decreasing or back, i.e. at the local maxima and minima of f' ($f''(x) = 0$ there): $x_1 = A$, $x_2 = B$, $x_3 = C$, $x_4 = D$.

x	$< x_1$	(x_1, x_2)	(x_2, x_3)	(x_3, x_4)	$> x_4$
f'	↗	↘	↗	↘	↗
f''	$+$	$-$	$+$	$-$	$+$
f	C.U.	C.D.	C.U.	C.D.	C.U.

The graph has inflection points at A, B, C and D.

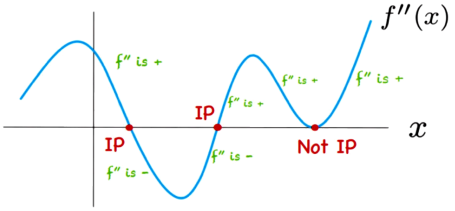


Ex 9 Possible inflection points are where $f'' = 0$: x_1, x_2, x_3.

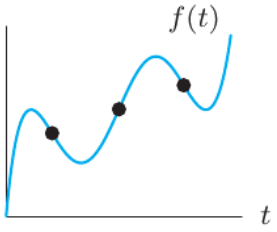
x	$< x_1$	(x_1, x_2)	(x_2, x_3)	$> x_3$
f''	$+$	$-$	$+$	$+$
f	C.U.	C.D.	C.U.	C.U.

f has inflection points at x_1 and x_2. x_3 is **not** an inflection point (f'' touches the axis but does not change sign).

The inflection points of f are where f'' changes sign.

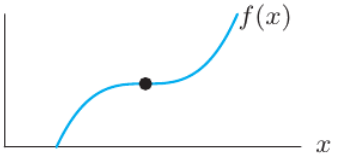


1 **Three**: one between the first local max and the local min, one between that local min and the second local max, and one between the second local max and the last local min (where the curve changes between C.D. and C.U.).



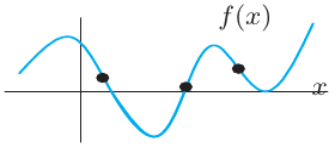
2 **Two**: the graph is C.U. at first, C.D. around the peak, then C.U. again as it levels off. One inflection point on each side of the peak (where the graph is steepest going up and steepest going down).

3 **One**: in the middle of the flat part, where the curve changes from C.D. to C.U.

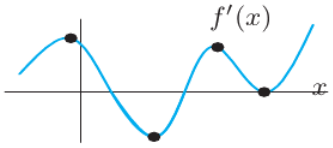


4 **Two**: one between the first local max and the local min (C.D. to C.U.), and one between the local min and the second local max (C.U. to C.D.).

35 (a) Where the concavity of the graph changes: one between each local max and the next local min (or min and next max): **3 points**.



(b) At the local maxima and minima of f' (where f' changes from increasing to decreasing or back): **4 points**.



(c) Where f'' changes sign (crosses the x-axis): **2 points**. The point where the graph only touches the axis is not an inflection point.

